

Required Readings

Required Readings for this program are open access, which means you can access them at no cost. The link provided in the citation should take you directly to the reading or to a page where you can download it.

1. [Board of Governors of The Federal Reserve System. *Financial Stability Report*. April 2024..](#)

- **Required pages:** Just pages i-vii and pages 1, 2, and 3
- **Estimated reading time:** 15-20 minutes

For some papers, it may be enough to read the overview or summary. This lesson is meant to reinforce that.

VALUATION IN NORMAL AND STRESSED MARKETS

Reading Time	90 minutes
Prior Knowledge	Comprehensive from Modules 1 through 6
Keywords	Bond Valuation, Stock Valuation, Option Valuation, ETFs, Risk

In the previous modules, we were introduced to several key themes. In this lesson, we will review the valuation of bonds, stocks, options, and ETFs in normal markets. Then, we will understand how these valuations change in light of stresses: changing rates, extreme events, and underestimated volatilities, as well as sector hits. In the case of bonds we devote a good amount of attention to the concepts of duration and convexity which are very important when working with fixed-income securities.

1. Bond Valuation in Normal Markets

Let's begin with bonds. Recall that bonds followed the discounted cash flows. Pricing a bond essentially involves finding the single yield, known as the yield to maturity, by which all cash flows are discounted. For bonds, those coupon sizes are fixed, hence the term fixed-income market, so this becomes a repeated exercise in discounting.

The industry throws an extra complexity in how it calculates time. There are day count conventions that measure how many days have elapsed since the last coupon was paid and how many days in total there are in a period. Not always intuitive, some of these day-count conventions do not use the actual number of days but rather a simplified counting system (e.g., 30 days in each and every month). For now, we will neglect day count conventions and instead focus on simple periods.

Suppose we have a 3% bond that has a 2-year maturity. In total, there are 5 cash flows. The yield-to-maturity is the number that makes the discounted cash flows match the market price. Maturity and coupon are known. Price and yield determine each other.

What if we don't know the size of the cash flow? Well, in the world of fixed income, that is not an issue. Indeed, this is where fixed income gets its name. These periodic payments are fixed in timing and size. (Let's not consider floating rates bonds here). The reliability of a fixed coupon makes bonds a relatively steady and safe investment. But what happens if we relax that assumption? In the next section, we'll cover stocks.

2. Stock Valuation in Normal Markets

We are now going to look at how to value a stock. There are several models to value a stock: discounted cash flow model, relative valuation, dividend discount model, etc. Here, we will consider the dividend discount model, which is predicated upon a company paying dividends.

Earlier you learned about growth stocks—that is not the subject here. Instead, we focus on value stocks, which are stocks that provide income. How do these compare to bonds? Well, bonds are required to pay their coupons or they go into default. Stocks are not required to pay a dividend, even value stocks, but historically have paid dividends consistently. If a value stock failed to pay a dividend, or even pays a lower-than-expected dividend, indeed, the stock price takes a hit. This is often why companies report their earnings and dividends after the market closes; otherwise, the sudden and surprising news could cause undesirable volatility. Clearly, then the dividend yield is crucial in valuing the stock.

What are the two key differences between a bond and a stock?

1. The bond has a specified, agreed-upon maturity date. At that date, the final coupon is paid and the notional amount reimbursed. After that date, no further cash flows occur. Stocks, on the other hand, do not have a "maturity" or an "expiration" date. Companies are theoretically meant to last forever. Therefore, the dividend stream of a stock is never supposed to end.
2. Second, bonds have a specified coupon size. Stocks do not. The size of the dividend is random. If a company does not earn profits in a given period of time and does not have retained earnings, or simply decides not to pay a dividend, there is no dividend. Therefore, dividend sizes are random variables.

To summarize, bonds have 2 known parameters: time to maturity and cash flow size. Stocks have neither of these. The problem of not having a time to maturity is solved by realizing the stock does not have one. The question of cash flow size is answered by modeling the size as a constant or variable. The variable itself can be modeled as one that grows at a constant rate or dynamic rate. We simplify the first by assuming there is no end period. How do we simplify the second? We may assume a constant dividend, a dividend that grows at a constant rate, or a dividend that grows at an increasing rate.

The constant dividend model is inflated if for no other reason than inflation. A company failing to pay a higher dividend is seen as not keeping up with inflation. If these were the model we used, then it would be very similar to fixed income valuation—except that we somehow have to handle the perpetual nature of the cash flows. When we value equities, we'll see that it pays to consider a dividend that grows at a constant rate or even one that grows at a variable rate.

3. Option Valuation in Normal Markets

Stocks and bonds are fundamental assets in the market. Derivatives are derived from these fundamental securities. When we value stocks and bonds, we are using cash flows. When we value derivatives, we tend to use replication arguments, which use stocks and bonds. Think of stocks and bonds as atoms. Yes, they have simpler parts (like protons and electrons). Think of derivatives as compounds containing atoms in different amounts. Indeed, it will be shown later in the program that an option can be thought of as a combination of a stock and a bond. You'll have to wait for Derivative Pricing to see the details, but for now, let's review the basics of option valuation in normal markets.

Suppose we want to value a call option whose underlying is a stock. We saw in a previous module that there were 6 key inputs. The first 5 of these are very easy to obtain from the market: the stock price, the strike level, the time to expiration, the risk-free rate, and the dividend yield. The final input, the volatility, is not as easy to observe. We need a model for it. One model we may use is the standard deviation of return over the last 30 days, or we may use a time-series model that recognizes volatility is mean-reverting and use that long-term (ergodic) volatility as the estimate for the volatility. Alternatively, we may simply use a proprietary model. When we put volatility in, we get a price.

Suppose we get volatility wrong. In this course, we saw that one problem with volatility is that it may not be constant. We examined distributions that we tested for normality, and then we showed the assumptions required for the normal distribution. Specifically, the kurtosis showed that the volatility itself seemed to have uncertainty with it. As a result, we may input a volatility whose value is less than the true volatility. In this sense, we underestimate volatility in normal markets. Nevertheless, a standard deviation may work well. That value becomes our input in pricing the option. In normal (usual) markets, hopefully, this volatility works well. But if we get this wrong, we underprice options: a costly mistake if we sold them and are hedging them to the tune of high volatility.

4. ETF Valuation in Normal Markets

A portfolio has combinations of securities. One way to estimate the return is to find the weights of each of the securities and the expected returns of each of the securities. Fortunately, return is linear, so we can simply take the weighted average of the returns. Using those returns, we can calculate a covariance matrix. The covariance matrix and historical returns can provide an expected measure of future volatility. If the covariance matrix refers to normal markets, then the covariance matrix should indicate future volatility. Of course, the markets may not remain normal but may become turbulent. To what extent would the covariance matrix degrade?

There are two ways the covariance matrix may change significantly. One is that variances get substantially higher. This would result in higher volatilities along the diagonal of the covariance matrix. Here, we have the same problem as in the previous section—the market may exhibit higher volatilities than we anticipated. Second, the correlations may increase, which means we have less diversification than expected.

We will have to wait until Econometrics to discuss issues like stationarity, which relate to variances becoming more or less constant over time.

5. Market Stress

5.1 Bond Markets

There are several things that can frazzle interest rates. In many countries, the central bank can set policy to raise or lower interest rates, in an effort to combat high unemployment and/or high inflation. Indeed, interest rate hikes and cuts can send other asset classes into a frenzy. For example, in the first half of 2022, not only interest rate hikes but also the acceleration at which those hikes were expected sent equities reeling, causing a stock market correction into bear territory.

Modeling interest rates by merely looking at a five- or 10-year history may not take into account older time periods that are more relevant to the central bank phases of expansion and tightening. More recent time series do not necessarily equate to more relevant data.

5.2 Equity Markets

In normal markets, the returns of stocks are shown to be fairly normal or Gaussian. In highly volatile markets, stock returns may still be Gaussian, but they may have much higher volatilities. Worse, the distributions may look more like a mixture of normals, with excess kurtosis and even jumps in the stock prices that increase the likelihood of an extreme event occurring.

5.3 Option Markets

In normal markets, options are modeled by careful calculation of the volatility of the underlying, the key input into pricing options in the first place. In turbulent markets, volatilities undoubtedly rise. Will we be able to price and hedge options correctly in a world where volatility reaches higher levels and rises and falls more quickly? Indeed, we can see volatilities easily exceed 100% for many equities and other asset classes in these markets.

5.4 Sector Markets

The issue of diversification that we thought was present may not actually be there. Suppose we are exposed to a sector that crashes. We may earn profits in multiple sectors but lose money in others. However, in highly volatile markets, correlations can rise, which lowers diversification. The promise of diversification to help hedge against a downturn turned out to be a false sense of security, as the heightened correlation treats the portfolio more like a single stock than a basket.

6. Interest Rate Risk with Duration and Convexity

By now, we are accustomed to associate volatility with risk when dealing with equities. However, bonds are different creatures. While volatility exists even for the price of bonds, bond investors are much more concerned about interest rate risk. And the reason is simple: when interest rate changes, so do the prices of bonds.

A key metric to gauge interest rate risk for fixed income investment is **duration**, which is a measure of the sensitivity of the price of a bond or other debt instrument to a change in interest rates. In general, the higher the duration, the more the price of a bond will drop as interest rates rise (thus we bear a greater the interest rate risk). For example, if rates were to rise 1%, a bond or bond fund with a five-year average duration would likely lose approximately 5% of its value. We will make this more rigorous in the following paragraphs.

NOTE. Making everything somewhat rigorous requires a bit of Calculus, but not much more than a knowledge of first and second order derivatives and Taylor's expansions. Something that is expected of students who plan to get a Master's degree in Financial Engineering. However, derivations/calculations will not be tested and only the use of these formulae will.

To understand how the duration and convexity formulae work, we start by recalling the formula for the price of a bond; i.e., the price of the bond at time t is given by:

$$P_t(r) = \sum_{i=1}^N \frac{CF_i}{(1+r)^{t_i-t}},$$

where CF_i is the cash flow (coupon payments and principal repayments) at time t_i for $i = 1, 2, \dots, N$, and r is the relevant yield to maturity or the market discount rate to use at time t . As you can see, $P_t(r)$ is a function of r and calculated at time t . Then, we can calculate the first and second derivative of $P_t(r)$ with respect to r , which with some tedious but simple calculations can be shown to be:

$$P'_t(r) = \frac{dP_t(r)}{dr} = - \sum_{i=1}^N \frac{CF_i}{(1+r)^{t_i-t+1}} (t_i - t)$$

and

$$P''_t(r) = \frac{d^2 P_t(r)}{dr^2} = \sum_{i=1}^N \frac{(t_i - t)(t_i - t + 1)CF_i}{(1+r)^{t_i-t+2}}.$$

These formulae may look cumbersome, but they only involve basic arithmetic manipulations of the kind easily performed using a spreadsheet. Alternatively, it is possible to code them in Python or any other programming language.

The formula involving the first derivative, $P'_t(r)$, is known as the **\$-duration** of the bond P_t at time t , when the yield to maturity is r . Instead, the second derivative, $P''_t(r)$, is known as the **\$-convexity** of the bond P_t at time t when the yield to maturity is r .

How do \$-duration and \$-convexity, play a role in helping us to assess interest rate risk?

Consider initially the case of a first order Taylor's expansion of $P_t(r)$:

$$dP_t(r) = P_t(r + dr) - P_t(r) = P'_t(r)dr + o(dr) \approx \frac{dP_t(r)}{dr}dr \equiv \$\text{-duration} \times dr$$

where $o(dr)$ is a notation to indicate that the additional terms not included go to zero when dr goes to zero, faster than dr does and thus are negligible. Thus, the change in value (dollars or another currency) is approximately equal to $P'_t(r)dr$ plus an infinitesimal term $o(r)$, which can be ignored for small movements in the yield to maturity, r . In other words, the \$-duration measures the change in the price of the bond due to the slope of the relationship between $P_t(r)$ and r , that is it informs an investor about how the price of the bond changes following a small change of the yield to maturity using a first order approximation.

Now, if we divide both sides of the previous equation by $P_t(r)$, we get

$$\frac{dP_t(r)}{P_t(r)} = \frac{P'_t(r)}{P_t(r)}dr + o(dr)$$

and, if we let

$$MD_{t,r} \equiv - \frac{P'_t(r)}{P_t(r)}$$

which called the **modified duration** of bond P_t , then

$MD_{t,r}$ provides the relative change in value in the price of a bond at time t for a small change in r ; i.e.,

$$\frac{dP_t(r)}{P_t(r)} \approx -MD_{t,r} \times dr.$$

NOTE. In the quizzes, you will just be given either MD or \$-duration or \$-convexity numbers and you will not have to be concerned about how they were derived. But this is a task often assigned to quantitative analysts and they inform us about interest rate risk for fixed income instruments like bonds.

Summarizing, it is useful to think in terms of profits and losses (\$&Ls); i.e.,

$$\text{absolute gain/loss} \approx \$\text{-duration} \times dr$$

and

$$\text{relative/percent gain/loss} \approx -MD \times dr.$$

NOTE. The last two formulae must be known to answer quizzes!

Let's see the formulae at work with a numerical example.

Example. Consider a bond with a maturity of 10 years, a coupon rate of 6%, and a yield to maturity of 5%. For simplicity, assume that the coupon frequency and compounding frequency are both annual. An analyst has calculated that the bond's \$-duration is -810 and the MD is 7.55. Assume that the yield to maturity decreases by 0.1%. What is the change in price for the bond in dollars and as a percentage?

Answer. The absolute loss/gain can be computed using the \$-duration. In this case, it is a gain and more specifically **absolute gain** = $-810 \times (-0.1\%) = 0.81$ i.e the price of the bonds goes up by \$0.81.

In percentage terms, the gain is

relative gain = $-7.55 \times (-0.1\%) = 0.755\%$.

Remark. Note that if the frequency of coupon payments or the compounding frequencies are not annual, the formula for dollar duration is different. It can (easily) be shown that in the semi-annual payments case we have:

$$P'_t(r) = - \sum_{i=1}^{2N} \frac{CF_i(t_i - t)}{(1 + r/2)^{2(t_i-t)-1}}.$$

However, we are not going there and leave it for the students to derive a general formula in the case of k -payments a year. Aspiring financial engineers need to be able to do that on their own.

As the relationship between the price of a bond P and interest, r , is non-linear, it is necessary to consider also the second derivative of the relationship. This is even truer if we consider changes in the yield to maturity that are not really quite small This leads us to need to include **convexity**. From a mathematical point of view, we can approximate $P_t(r)$ using a second order Taylor's expansion:

$$dP_t(r) = P'_t(r)dr + \frac{1}{2}P''_t(r)(dr)^2 + o((dr)^2)$$

and, recalling what we did a little earlier, we can write this formula as:

$$dP_t(r) \approx (\$ - \text{duration})_{t,r} \times dr + \frac{1}{2}(\$ - \text{convexity})_{t,r} \times (dr)^2$$

where this time, we are including small terms like $(dr)^2$, but we are ignoring terms of order $o((dr)^2)$, which we consider negligible for our purposes.

$$(\$ - \text{convexity})_{t,r} \times (dr)^2$$

measures the change in the price of the bond due to the curvature of the relationship between $P_t(r)$ and r .

The last formula above, could be easier to remember as follows:

$$\text{absolute gain/loss} \approx \$\text{-duration} \times dr + \frac{1}{2}\$ - \text{convexity} \times (dr)^2.$$

That's enough math for us in this course. Let's review what we learned regarding bonds.

- When the relevant interest rate increases, then the price of the bond decreases. If the interest rate decreases, the price of the bond increases.
- The amount by which the price increases or decreases can be determined using \$-duration (first order approximation) or \$-duration and \$-convexity (second order approximation).
- The formulae provide accurate results only if the change in the interest rates are small. If the change in interest rate is not small, say in the order of a few basis points, then the formulae are in general not accurate and, in any case, using convexity and duration together would provide more accurate approximations.
- It should also be clear that the higher the \$-duration is, the more sensitive to changes in interest rates the price of the bond is and the more important it becomes to hedge interest rate risk.
- The change in interest rate are assumed to happen in a parallel fashion, i.e. the entire yield curve moves up or down by the same amount (parallel shifts). However, sometimes, the yield curve steepens, flattens, or twists. These non-parallel movements are not handled well by traditional methods of duration. Other methods, including principal components, can be used to provide risk measures of how much variation is caused by tilt, twist, and other types of yield curve movements.

Duration as a concept is more than a century old! It predates computers, spreadsheets, and software, all of which today can easily compute the risks associated with movements of the yield curve that are not parallel. While duration still has its use, when we want to understand the effects that stressed markets can have on bond prices, and fixed income assets in general, we can use duration and other measures to examine the sensitivity of prices to changes in yield.

7. Equity Risk with Heavy Tails

In a healthy market, volatility of a stock would remain constant. Then, its corresponding options can be priced with a constant volatility as if the equity's returns are Gaussian. Of course, markets do not always follow the assumptions we would like them to have. Therefore, the volatility can change in stochastic ways. We may want to treat volatility as a parameter—a constant. But markets tend to show that it behaves like a variable—a time-varying quantity. Variables can produce extreme values, and this gives rise to heavy tails. Perhaps you will hear someone remark, "Wow, this was a big loss for stocks, a 3-sigma move—such an unlikely move." Keep in mind that they are providing a perspective that views the world as Gaussian. Indeed, in the Gaussian world, 3-sigma events are indeed rare: perhaps we get three events every four years! (99.7% of the time, data lies within three standard deviations. That means only .3% of events are outside that interval. With 250 trading days a year, that equates to about three events every four years). Yet, who says that equity returns have to follow Gaussian distributions? At best, they may be Gaussian, but the volatility itself is changing; that is, we underestimate volatility. At worst, they are a more complicated structure, such as non-Gaussian, or a mixture of normal random variables, or even a time-changing distribution that requires regime-switching models or other complicated modeling.

As you read this lesson's required reading "Financial Stability Report," you'll see that asset valuation depends very much on the compensation investors demand for taking risk—the so-called risk premium. At times, the appetite for risk is so great that there are high asset valuations; that is, stocks are expensive when comparing their price to earnings or similar metrics. In stressed times, that appetite can drop significantly and abruptly, indicating a very different price investors are willing to pay for risk. As the paper indicates, "The risk premium for an asset varies over time and, unlike the price of an asset, cannot be directly observed." This is just one of many reasons why equity valuation is among the most challenging problems in finance, and perhaps one of the areas with the greatest amount of disagreement. Further discussion of asset valuation can be found in the semi-annual report published by the Federal Reserve.

8. Option Risk with Underestimated Volatility

Following up on the previous section, difficulties in modeling volatility lead to difficulty in pricing options. Volatility could be greater than we think for an untold number of reasons. Let's consider several of them.

First, in a Gaussian world, options can be priced with a constant volatility. We just saw that if volatility is a variable rather than a parameter, we should expect volatility to change over time. So what if we based volatility on a relatively low period of volatility, but then volatility is higher? In this new volatility world, returns could still be Gaussian, but we've underestimated the volatility: a statistical error of inferring too small a value for the statistical estimate.

Second, we may have incorrectly assumed that the stock returns were Gaussian in the first place due to heavy tails: kurtosis. Suppose their distribution is closer to a Student's t-distribution. Then, we would tend to underestimate the frequency of large moves by observing the world with our Gaussian glasses.

Third, we may have incorrectly assumed that the stock returns were Gaussian because we assumed symmetry when in fact there are skewed tails indicating asymmetry. Then, we would tend to underestimate the frequency of a negative return because we didn't take this skewness into account.

Fourth, we may have incorrectly assumed that the stock prices could not jump and therefore assumed that extreme moves could only occur if "3-sigma" or "4-sigma" or similar extreme events occur. The Gaussian distribution, however, shows that these types of events are extremely rare.

Fifth, we may have underestimated the costs and complexities of hedging the position, which adds costs to the volatility itself. In other words, real-world costs of hedging make the implied volatility higher than realized volatilities. That makes option pricing risky and hedging more difficult. At worst, we underestimate the size of volatility and the costs of hedging and sell options into the market too cheaply. Then, the options go into the money, and we find ourselves hedging at considerable expense.

Prudent option writing and buying require us to have a solid grasp of both the complexities of volatility and the dynamics of hedging. For example, hedging a put may require the option writer to sell short the underlying stock—a transaction that requires a financing cost. That financing cost can vary over time and get prohibitively expensive. So not only is there a risk of underestimating volatility, but there is also a risk that we underestimate the cost of hedging.

9. Sector Risk with Factor Exposure

One takeaway from COVID-19 is how fragile the supply chain can be. For example, an automobile can have 99% of its parts readily available, but one of them—perhaps even the microchips—are unavailable due to disruptions in the supply chain. These may arise from the inability to source the materials, new regulation that prohibits importing materials from countries that are not considered "free-trade counterparties," the lack of shipping and trucking to transport goods, etc. The consequences are far-reaching (The Board of the International Organization of Securities Commissions). There can be inadequate supply. In addition, the demand can be pent up—during COVID-19, there was greater demand worldwide for automobiles as people reduced their use of public transportation. You can see how a sector like automobiles can be impacted.

On the other side, business travel was greatly reduced during COVID-19, from regular business travelers to conference attendees. Remote work has become increasingly acceptable for many businesses worldwide. Of course, there are still many people who want to take vacations (holidays) and can still choose to travel, but disruptions to worldwide travel significantly impact the airline and hotel industries.

Sectors can be impacted by a variety of factors, including global pandemics. New regulation, geopolitical risk, and inflation affect sectors and can boost or bust a sector. The classical methods of examining which sectors perform well in specific business cycles need updating to provide better analytics. The key is often to find the underlying factors that affect sectors and test their vulnerabilities.

Remember, though, that good ETFs have strength in numbers: diversification by having pooled idiosyncratic risks. ETFs were shown to be very resilient during the COVID-19 pandemic. See <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD682.pdf>

10. Conclusion

Bonds, equities, options, and ETFs are four of the asset classes we've examined in this course. This introductory lesson compared how their pricing and sensitivity change normal markets compared to stressed markets. In the next lesson, we'll tie these in with the sets of related financial concepts.

References

Board of Governors of the Federal Reserve System. *Federal Reserve Financial Stability Report*. November 2022.

The Board of the International Organization of Securities Commissions. *Exchange Traded Funds Thematic Note - Findings and Observations during COVID-19 Induced Market Stresses*. OICV-IOSCO Public Report, August 2021. <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD682.pdf>.

KEY CONCEPTS IN FINANCIAL MARKETS

Reading Time	90 minutes
Priority Knowledge	Comprehensive Modules 1 thru 6

In the previous lesson, we compared how securities get priced in normal markets and then the challenges that throw these methods off course. In this and the next lesson, we review the first 10 sets of key financial concepts: collateral, statistical, magnifying, frictional, and failure. These concepts affect the data we use, the models we build, and the decisions we make when we are solving problems that involve pricing, hedging, and allocating funds. We will review these concepts in pairs and see some examples that combine several concepts at once.

1. Collateral-Related Concepts: Credit Risk and Financing

1.1 Long Trades vs. Short Trades

Remember that in financial markets, investors can enter a trade in two distinct ways. They can purchase a security they do not own—this is known as *going long*—or they can borrow a security they do not own from someone who does own it and sell it—this is known as *going short*.

- When the investor is long, they anticipate the price of the security increases so that they can sell the security at that higher price. When they do so, they close the position and find themselves with a profit as the net result of the two transactions. The plan is to *"buy low, then sell high."*
- When the investor is short, they anticipate that the price of the security will decrease. The investor hopes that they can buy the security at a lower price later. When they buy it back, known as *"covering the short,"* they receive the security, return it to the party that loaned it to them, and close their position. The plan is *"sell high, then buy it back low."*

What does an investor need in the situation where they go long? In a word, funds. If this is an individual, they need to have cash in their account. If this is an asset manager, they need to have the funds of their clients. Of course, what if they do not have enough funds? They can finance by borrowing money, paying for a portion of the securities with borrowed funds and the rest with actual funds. Long positions have the flexibility of using financing or not, depending on whether the investor wants to be more conservative (using all cash) or invest more aggressively (using financing/leveraging).

What does an investor need in the situation where they are short? They need to borrow the security. Who loans it to them? Someone who owns that security! Well, why do you think someone would loan a security to another investor with whom they neither met nor have financial relationships? The answer is income: the owner of the security will be paid a financing fee. Thus, financing is not free. Anything borrowed has an associated cost. This cost is assumed to be different prices in different models. In Black-Scholes, this cost is assumed to be zero. That is not a great assumption. In a paper by the late quant Marco Avellaneda called "Option Pricing for Stocks that are Hard-to-Borrow," the cost is assumed to be a Brownian motion that can vary with the supply and demand forces for that security. This model for financing costs is more realistic and can lead to better pricing in the options market.

2. Credit Risk and Financing

Credit risk and financing are the first set of challenges we address in financial engineering. Credit risk has been defined several times in different places in our course. Think of credit risk in simple terms: a lender gets less money back than what was originally lent or expected back. Recall from the first few lessons you saw credit risk from the bank's side (in a mortgage) and from the depositor's side (with the lender). In 2022, the failure of FTX to secure funds of investors illustrates the credit risk that exists when funds from brokerage accounts are lent out and not returned. The party that is owed money likes to keep track of their counterparty's credit risk. When collateral is in place, the problem extends to financing and a calculation of market risk (of that collateral). We discussed how collateral, regulation, and insurances (like the FDIC) help to mitigate credit risk.

In subsequent modules, you read that lending rates are a function of several key factors, three of which include the term of the loan, the creditworthiness of the borrower, and the type of securitization, if any, tied to the loan. During markets where the yield curve is upward sloping, the rates for short-term loans tend to be lower than the rates for long-term loans. The rates for securitized loans are lower than the rates for un-securitized loans. For example, home mortgage interest rates are lower than consumer credit card rates. If a homeowner defaults, the bank can take the home as a substitution. If a credit card borrower defaults, the bank may have nothing tangible left—the funds could have been spent on massages, products, and services that were consumed or have lost significant value. Finally, the rates for better creditworthy individuals are more favorable than those borrowers that are considered high risk.

2.1 Examples of Credit Risk

The ability to model credit risk allows the financial engineer to solve a vast set of problems. How do you price a corporate bond? Given a borrower's credit history, what is a suitable interest rate for a securitized loan? What would BBB bonds expect to recover if they were to default? These questions require us to understand the key variables of credit risk.

Let's take the corporate bond example. We'll start by pricing a bond that cannot default and then, somehow, include the risk of default. In credit risk, we have various ways that you can measure the risk of default. This is known as the probability of default. We might get there by looking at stocks, credit default swaps, credit ratings, or other factors we build. Therefore, we need financial data and models. For example, sometimes, we'll use the capital structure of a company to model default probability. We'll want to know the company's assets and debts. Then, we'll model its equities like an option. You saw that in Module 4. We'll expand upon it in our discussion of options.

Credit risk is influenced by, and influences, asset classes during times of market stress. Higher default probabilities or lower recovery rates are two examples of credit risks that can increase. Higher default probabilities will increase funding costs, raise debt-to-equity ratios (discussed in Module 5), force higher collaterals to be set aside as cushions for unexpected losses, and make it more difficult for a firm to be profitable. As markets get more stressed, this in turn nudges higher defaults, lower recovery rates, and greater expected losses. For example, consider a borrower with a variable-rate mortgage. Suppose interest rates increase. In turn, this will likely increase mortgage rates. With a variable-rate mortgage, the costs of the mortgage have increased. Suppose the borrower is no longer able to afford these higher payments. At the same time, higher interest rates and mortgage rates also caused home prices to drop, as the higher rates decrease the demand for housing. The value of the collateral—the house—is now worth less! So there are two credit risks that worsened: the default probability increases and the loss given default increased, since the collateral is now worth less.

In credit risk, there's several variables involved in the expected loss: default probability, loss given Default, and the exposure at default. The first two of these are correlated. We're not escaping correlation. We're just doing one set of factors each week. But you can see how that can lead to what's known as different ways that you would model an expected loss.

2.2 Building the Credit Risk Toolkit

Credit risk has real-world practicality. Personally, you may decide you want to lend a friend money. Somehow, you may run a quick model in your head: "Have I already lent them money? Have they paid me back? Are they likely going to pay me back? Are the funds used for something serious?" Credit risk is something that most people can understand intuitively.

So what do we do with credit risk?

- In Econometrics, we'll look at regression models, such as logistic regressions, where we can model a default as a Poisson process. We will get exposed to different distributions that are suitable to model default probability and loss.
- In Derivative Pricing, we'll understand default as a probability like a European call option ending up out-of-the-money. We'll also learn new products like the credit default swap that isolates credit risk, making it different than the econometric approach.
- In Stochastic Models, we'll examine examples of calibrating credit default swaps to imply a default probability. We'll expand our mathematical toolkit to include Markov processes to allow us to model the migration of credit transition matrices.
- In Machine Learning, we'll use supervised learning to improve upon models that predict default.
- In Deep Learning, we'll provide more detailed ways we can back into credit risk variables from observed credit derivative products like first-to-default baskets.
- In Portfolio Management, we will measure and manage credit risk in examples where we use copulas to model joint defaults.
- In Risk Management, we'll use a network approach in Bayesian networks to model credit risks.

Clearly, credit risk is a recurring theme throughout the whole program. There is no escaping credit risk, particularly since the global financial crisis when so many experts got credit risk wrong. Since the GFC, even something as simple as pricing an interest rate swap now requires credit risk calculations, known as XVA adjustments, that credit or debit value based on the credit risk of you and your counterparty's credit risk.

3. Financing

3.1 Examples of Financing

Financing reflects the cost of borrowing something. Clearly, it is related to credit risk. Suppose you short a security. Step one is that you borrow it from them. The owner charges a fee—that's where financing enters. Owners of securities who lend them out do so to make extra return. Let's say that you owned Meta stock (formerly known as Facebook). You agree that you are willing to lend your shares to earn some extra income (income in addition to any dividends it would pay). Other people may be bearish about the stock and believe it's overpriced, so they may want to sell the stock. But they currently don't own it. They can borrow shares from you. You would then charge whatever the market rate is, let's say it's twenty basis points (meaning 0.20% interest on an annualized basis). Once they have the stock, they sell it short in the market. They're hoping the stock price drops, so they buy it back (cover) at a lower price. If that becomes true, then they've made money by selling high and buying low. However, the price could increase, and they may have to cover at a higher price.

Either way, the borrower had to pay a financing cost to the original owner. Securities are not library books. You could go to your local library and presumably borrow a book for free and return it two weeks later. That's not how securities work. If you went to a brokerage and said, "I want to borrow shares of Meta," whoever owns those shares would agree when they open their account to lend those securities, and that would earn them extra income. Suppose Meta paid a dividend. Then, the borrower who borrowed the stock also has to pay the dividend in addition to the financing cost. The true cost of borrowing is a combination of the income it produces (that is, the dividend) and the financing cost, volatile as it may be.

3.2 Building the Financing Toolkit

- In Econometrics, we'll see how the cost of carry can be modeled as a time series that depends on certain exogenous variables.
- In Derivative Pricing, we'll look at a very simple financing model (hint: no cost) and how you hedge a put by shorting the underlying with the Black-Scholes model. However, if we just used these assumptions, we would never reproduce the volatility smile.
- In Stochastic Models, we'll look at a model of how you can include financing costs in the hedge. For example, we'll model the cost of financing as an additional Brownian motion and show how this model reproduces the real-world option prices better.
- In Machine Learning, we'll examine financing associated with performing a long-short strategy, where you buy one security and sell short another, with the objective that the long security outperforms the short security.
- In Deep Learning, we'll extend some previous examples to use more market data.
- In Portfolio Management, we'll use financing costs as part of constraints in minimizing costs to rebalance a portfolio.
- In Risk Management, we'll discuss liquidity crises that affect financing costs.

4. Understanding the Motivation of the Curriculum

Before we look at the next set, fast forward two years from now. As you're worked through the program, a few of these factors have become your favorite problems. For example, if you enjoy solving problems in credit risk, you could certainly find work in that area. Within the field of credit risk, there are specializations: counterparty credit risk modeling; credit rating agency factor modeling; credit derivative trading; corporate bond portfolio management; structured credit sales; credit risk management; and so on. When you're getting assessed later in an interview, think about the tools you learned. Rather than thinking, "what did I study in Econometrics?" you can ask yourself "what skills did I build in Econometrics that I applied to solve credit risk problems?"

So you may wonder, why not have one course dedicated to each of these areas, like credit risk and volatility modeling, instead of courses like Econometrics and Machine Learning? This relates to the difference between theory and practice. Econometrics has foundations in linear algebra, probability, statistics, and calculus. These are used to build models. Once you have these statistical, mathematical, and machine learning models, you can apply them in many ways to solve problems in credit risk, financing, leverage, regulation, etc. The educational process in engineering could not possibly teach you every application. That's its strength and beauty. Instead, we apply the models to solve problems in these areas of estimating credit losses, measuring volatilities, and more. Some courses will emphasize a particular factor. For example, Derivative Pricing will illustrate the power of both leverage and non-linearity. Others can be broadly applied to any areas. Machine Learning can tackle problems across the spectrum.

You are encouraged to think of each metric, methodology, or model as a tool, as if you were a carpenter. Imagine you are training to be a carpenter. You're going to develop skills in modeling, measuring, cutting, adhering, smoothing, etc. To perform these skills, you would learn to use a variety of tools. Likewise, think of these areas as opportunities for skills development: skills for handling credit risk, addressing financing, etc. As you work through the curriculum, think about how the material illustrates how a tool is used, teaches you to use that skill, and builds a foundation to understand its justification. Don't miss the forest for the trees! Instead of overemphasizing technicalities, focus on the big picture. Ask yourself, "how did this course improve my mathematical, statistical, or computational background to better understand, model, predict, or interpret credit risk? Volatility? etc."

5. Statistical Factor 1: Volatility

A big part of the Financial Markets course addressed volatility and correlation. Volatility means that we have uncertainty around an expected value. We expect a certain return, a certain payoff, a set of cash flows, etc., but we often get different results. Volatility describes that expected difference.

Considering more than one security at a time, we have a covariance—how two or more securities vary together. This depends not only on their individual variances but also on their correlation. Correlation is the degree to which two securities move together. Of all these financial challenges, most of the attention in your studies will focus on ways to measure and manage volatility and correlation. Let's define volatility and then discuss a dozen of its properties.

- Volatility can mean risk or uncertainty.
- Volatility is indifferent to direction.
- Volatility is nonlinear.
- There are different types of volatilities.
- There are different ways to measure volatility.
- Securitized portfolios have securities with similar variances.
- Volatilities change over time.
- Volatilities do not necessarily scale with time.
- Volatilities can be implied from derivatives, especially volatility derivatives.
- Volatilities affect other variables.
- Variances are the diagonal of the covariance matrix.

5.1 Volatility can mean risk or uncertainty.

Volatility is often used synonymously with risk, a term people use often: risk management, risk aversion, risk premium. Risk typically describes situations where the outcomes—their distributions and probabilities—are known. Let's say we're flipping a coin many times. If all the events are independent and we run lots of them, the long-term results should converge to a normal distribution. Then, we could characterize the risk through metrics like Z scores, which rely on standard deviations. The calculations and methods you know and use from statistics find a home when discussing risk. Although we don't know what a specific outcome is, we have general comfort in knowing what the distribution is. When you studied statistics, you learned of these different distributions. Some were Gaussian. Others were skewed or kurtotic. That is, many have heavy tails compared to the normal distribution. The heavier the tails, the greater the chance of an extreme outcome (assuming we don't have disproportionately more chances of zero returns). In simpler terms, volatility is more uncertain in heavy-tailed distributions. For example, in 2022 Q4, the New York Stock Exchange set a record: the market moved down and then up so much that the total distance covered was more than ever before. Although there have been larger moves in one direction on a given day, the record was the comeback within a single day. This reflects very high intra-day volatility. It means that the distribution that we have may not be Gaussian.

One thing that makes volatility difficult is distinguishing risk (known distributions) from uncertainty (unknown distributions and/or outcomes). How do we know that we have the right distribution? If we had some understanding of the shape and the value of the parameters, then we're using classical or Bayesian statistics to draw inferences. For example, parametric Value-at-Risk (VaR) requires a distributional assumption; then, we pick a percentile and report that losses do not exceed a specific value, say, 99% of the time. When we are unsure of the distribution, or even the outcomes and probabilities, we switch the terminology from risk to uncertainty. Perhaps the return of Bitcoin falls into this category. Uncertainty means there's less consensus (if any) on the underlying distribution(s). You may have unstable parameters, changing shapes, or mixtures of distributions. For its own name, you have a distribution known as a mixture of normals—a combination of 2 (or more) normals, each with its own mean and variance, plus a proportion of how they mix. Perhaps the second normal has the same mean, but a much larger standard deviation. You put these together, and you have a mixture of normal distributions that introduces some of the heavy-tailed nature that you'll see. The parameters of the mixture (for the bivariate normal there are 5) may be time-varying.

Risk and uncertainty describe what you'll see in this whole master's class as volatility. In the Financial Markets course, Modules 2 and 3 presented to you two different asset classes. You saw both cryptocurrencies and stocks. Those were chosen because those asset classes are a lot more volatile than others. Even within stocks, you have some stocks that are much riskier than others because they are companies that are younger or perhaps just belong to industries that inherently have more uncertainty.

5.2 Volatility is indifferent to direction and semi-deviation.

Indifference to direction can be interpreted by saying that if the order of the elements is changed, their standard deviation does not change. In other words, we could organize the data in ascending or descending order and the value of the volatility would not change. This uncertainty is indifferent to direction. When risk increases, there are greater chances of moves in either direction. Volatility relates to the size of a move rather than its sign. Consider a random variable that is bounded; for example, default probabilities lie between 0 and 1 inclusive. Suppose the current value is 1%. Increasing volatility will mean it can move in either direction, but to be sure the boundary at 0 means there is much less room to move on the downside than on the upside.

However, we could also interpret *"indifference to direction"* as a lack of distinction between "good" and "bad" volatility. For instance, assume that an investor buys shares of XYZ at a price of \$50/share. Then, suppose that the price keeps going up almost daily and that 1 year later the shares of XYZ are valued at \$120/share. The stock has exhibited a good amount of volatility during the last 12 months, but none of the investors who bought the shares of XYZ would be upset. On the other hand, if the stock kept going down and 1 year later it stood at, say \$15 per share, investors in XYZ would be very upset. Volatility does not distinguish the good direction from the bad one and it were less to assess risk from the volatility alone, we might conclude that the investment was riskier when it went from \$50/share to \$120/share than it was when it went from \$50/share to \$15/share. Clearly, that does not seem to cut it.

For this reason, portfolio managers, besides regular volatility, may compute what is known as **semi-deviation**. The downside semi-deviation measures the volatility or dispersion of negative returns below a specified threshold. Let $\mathbf{B}$ represent the desired target return; the sample target semi-deviation formula is given by

$$s_B = \sqrt{\frac{\sum_{r_i \leq B} (r_i - B)^2}{n - 1}}$$

where $\mathbf{N}$ is still the number of observations in our sample, but the sum considers only those returns, r_i 's that are below the required threshold $\mathbf{B}$.

Example. Fund MNO over the last 10 years have achieved the following returns:

4.5%, 6.0%, 2.2%, -2.0%, 0.0%, 5.2%, 4.1%, 2.5%, 5.5%, 4.3%.

- Compute the standard deviation of returns for fund MNO over the given 10 years;
- Compute the semi-deviation of returns over the same period for fund MNO if the target return is 2%.

Answer. The standard deviation can be computed using the usual formula for the sample standard deviation and it is equal to 1.57%.

To compute the required semi-deviation, we need to use the formula given above with $\mathbf{B} = 2\%$, and $\mathbf{N} = 10$. Then, since only the returns less than or equal to 2%, matter, we find

$$s_B = \sqrt{\frac{1}{9} [(-2.0\% - 2.0\%)^2 + (0\% - 2\%)^2]} = 1.49\%.$$

The notion of semi-deviation just introduced focuses on the volatility of returns below a specified threshold or target return and allows us to better estimate the likelihood of experiencing negative outcomes.

5.3 Volatility is non-linear.

Volatility is non-linear. If we were to combine the volatilities of 2 securities, it is not simply their weighted average. Correlation affects the total. We'll see this explained in a few points.

5.4 There are different types of volatility.

Sometimes, the word volatility will have another term in front of it. You will see historical volatility, realized volatility, implied volatility, local volatility, stochastic volatility. Some of the distinction is due to the securities used to calculate the volatility. Other distinctions are based on the type of model used.

5.5 There are different ways to measure volatility.

Volatility can be measured with different metrics: variance, semi-variance, inter-quartile range. It may use different data that results in different metrics. For example, using prices we can compute the price range, high price minus low price. Using returns, we can compute variances.

5.6 Securitized portfolios have securities with similar variances.

Recall in discussions of products like MBSs, the securities should have similar risk characteristics. This would facilitate modeling their prepayments. Imagine if the mortgages had very different maturities, interest rates, types (fixed or variable). It would be incredibly difficult to model the overall MBS prepayment. Securitization relies on having securities with similar risk profiles.

5.7 Volatilities change over time.

Sometimes, we'll assume volatilities are constant. (See the Black-Scholes model.) As we develop stronger modeling skills, we'll assume volatilities change. GARCH, by definition, models a changing variance: the H represents heteroskedasticity, from the Greek words for "different variances." In modeling option prices, stochastic differential equations will provide more flexibility to include variance as an additional source of variation aside from the underlying. One of the key choices we'll make in any model is which quantities are parameters versus which ones are variables.

5.8 Volatilities do not necessarily scale with time.

Volatility depends on the timescale on which it was measured. One thing that you really need to understand early and appreciate as you start collecting data and think about models is at what frequency you're measuring data. Most of who you will start by looking at daily closes: take the end-of-day price. That's well defined because markets have an official closing price. Some markets trade 24/7 (e.g., Bitcoin), so there is no closing time. Perhaps you trade high-frequency and need to examine volatility within minutes or seconds. The key point here is that volatility over, say, 1 minute, is a neat fraction of the volatility for a longer period. Sometimes, you may assume volatility scales like a Gaussian, but this is often not advised.

5.9 Volatilities can be implied from derivatives, especially volatility derivatives.

Options imply volatilities. There are different types of derivatives: vanilla prices, variance swaps, volatility swaps, etc. Through calibration techniques you will learn later, you can see how market prices imply volatilities.

5.10 Volatilities affect other variables.

Higher volatility—due to uncertainty in the market—affects liquidity. Similarly, diminishing liquidity can affect volatility. They affect each other in complex ways. The same can be said for most of the other financial challenges.

5.11 Variances are the diagonal of the covariance matrix.

Within a covariance matrix, the diagonal element contains the individual variances of the interesting variables. So securities that are more volatile will have higher entries. The correlation matrix is scaled, so that can be a more interesting way to see relationships on a more normalized basis.

In short, more volatility means we can expect to be further away from an expected outcome or perhaps even an outcome we never imagined.

6. Statistical Factor 2: Correlation

Let's emphasize some stylized facts about correlation, specifically things that are in the notes for this module and that relate to the previous two modules things. You'll also see these in upcoming modules and even upcoming courses.

Let's consider a dozen properties that are useful to know about correlation from the onset.

- Correlation is bounded and does not have units.
- Correlation does not have a direction.
- Correlation measures linear or monotonic association.
- There are different types of correlation.
- There are different ways to measure correlation.
- Correlation does not imply causality.
- Good regressions have low correlation between predictors.
- Diversified portfolios have low correlation between securities.
- Correlations change over time, especially when stressed.
- Correlation affects over variables (e.g., wrong-way exposure).
- There are alternatives to correlation.

6.1 Correlation is bounded and does not have units.

Correlation is bounded between -1 and 1. Most asset correlations are positive. Extreme correlations (-1 or +1) are rare in practice.

6.2 Correlation does not have a direction.

Correlation does not have a direction. This means that we can swap the order of terms. For example, the correlation of net income and profitability is the same as the correlation of profitability and net income. Note that this is not true for regression. The regression of net sale on probability is a different model than the regression of profitability on net income. Regression is directional; correlation is not.

6.3 Correlation measures linear association.

Pearson correlation measures linear association. Recall in our discussion of correlation we also studied Spearman correlation. Spearman measures directional association. Suppose you have two variables, x and y, and they are related through a CAPM-like linear relationship. Graphing y versus x shows a linear relationship. If you switched the relationship to a quadratic, where y contains a squared term in x and a linear term in x, the correlation is not as linear. The higher-order term introduced non-linearity, which would decrease the correlation. You can still have as strong of a relationship, when you include the x-squared term; it is just not linear.

There are examples of nonlinear relationships we have already seen. Recall the graph of a bond's price versus its yield. We get an inverse but convex relationship: the higher the price, the lower the yield. The Pearson correlation of price and yield is not minus one because of convexity. Compare that to the Pearson correlation. Think of Spearman as the direction rather than the size of the move. The Spearman correlation of price and yield is a negative one. In other words, if we increase the price, we lower the yield; we don't do it in a linear way. When the yield is high, small changes in yield cause relatively small changes in price. When the yield is low, small changes in yield cause relatively big changes in price. Those price-yield sensitivities are a result of convexity. When the yield increases, the price increases. Therefore, the ranks of the price and the ranks of the prices are negatively correlated. Recognize the difference between the Pearson and Spearman correlations: Spearman removes the linear part of correlation and focuses on direction.

6.4 There are different types of correlation.

Depending on the variables we compare, we can calculate different types of correlations. Using two sets of stock returns, we can compute equity correlations. Using two sets of yield, we can compute yield correlations. Using two sets of default probabilities, we can compute default correlations. In credit risk models, we may compute asset correlations.

6.5 There are different ways to measure correlation.

Classical correlation is done using Pearson correlation. There are non-parametric versions like Spearman correlation and Kendall correlation. For time series, we compute autocorrelations and partial correlations. In your studies, be sure to pay attention to any word in front of correlation to understand the specific type of correlation being calculated.

6.6 Correlation does not imply causality.

When we see correlation, we should not assume causality. We don't want to assume when we see correlated events that one causes the other. There may be confounding variables. Let's take a classic example. Suppose you find that a child's reading score and shoe size are correlated. Would you say that having bigger feet improves reading ability? The confounding variable here is age. Age and shoe size are correlated; as kids age, their feet grow. Age and reading ability are correlated; as children learn more, they improve their reading. Sometimes, when examining economic variables this is not so obvious because these variables like interest rates, inflation rates, and unemployment rates affect each other in complex ways. There's not a clear cause-and-effect relationship in observational markets. Finance and economics are physical sciences where we are better suited to conduct controlled experiments. Remember, from property one, correlation is not directional, so we can't get a good sense of what comes first.

6.7 Good regressions have low correlation between predictors.

When you use multiple predictors or factors in regression, you don't want them to be correlated! If you run a linear regression, you don't want to use two factors that are highly correlated with each other. Suppose you're trying to predict the price or return of a company. Say that you include both the market cap and the market share. Imagine for the selection of stocks in your study, these two variables are highly correlated. The predictors are bumping into each other. You don't want to have too high of a correlation of your predictors. They don't have to be a zero correlation, but you don't want one too high; otherwise, you create problems in the regression. The problem is a violation of one of the elimination assumptions about residuals being independent of each other. In some examples you'll see, we'll completely ignore this problem when we use principal components to transform our data into uncorrelated factors. This is a reason to upgrade your math skills so that we're not just measuring correlation.

6.8 Low correlation enhances diversification.

Fifth, just as in our regressors, we prefer to have relatively low correlation in our portfolio. This low correlation facilitates diversification. If you go to portfolio examples well before the portfolio management course. For now, think that when there is perfect correlation, there is absolutely no diversification. Imagine a vertical line stretching from 1 to -1. Correlation can take any value along that line. As you move down from one to minus one, correlation decreases and diversification increases. If you had a correlation of point nine, you'd still have some diversification, but not much. Recall from Module 3 how diversification benefit is defined. We had an equally weighted portfolio of two stocks, one with a standard deviation of 3 and one with a standard deviation of 8. When the correlation was minus 1, the overall standard deviation was less than the standard deviation of the "safer" one: -2.5. As the correlation increased, the standard deviation increased in a non-linear (i.e., with curvature) way. When the correlation was 1, the standard deviation was just a straightforward average of the two securities. Clearly, as correlation decreases, the diversification benefit increases because the standard deviation decreases. The counterintuitive part is that there is a correlation that makes a combination of the two less risky than either individual security. That's the power of diversification. Incidentally, we call a correlation of -1 a perfect hedge. However, a perfect hedge is not necessarily desirable since it would protect unwanted outcomes but at the cost of wiping out any chance of profits.

Note also you can perform these calculations in your head. The securities have standard deviations of 3 and 8.

With perfect negative correlation, one offsets the other. The difference between eight and three is five: split five in half (with securities equally weighted) and you get a 2.5 standard deviation—remarkably, lower than either security.

With perfect positive correlation, their risks are additive. The sum is 11, making their average 5.5. That's the standard deviation: 5.5.

Due to the relationship between standard deviation and correlation being nonlinear, that's all the calculations you can do in your head. If the correlation is 0—centered between -1 and +1—the standard deviation is NOT 4.0, the average of 2.5 and 5.5. That is because it is nonlinear.

The fact that a combination of securities can be less risky than either security individually, even if both are held long (and not short), is a common interview question: "Can a 2-stock portfolio be less risky than either security in it?" The answer is "Yes, depending on the correlation!" At minus one, that's possible.

6.8 Correlations change over time.

Like volatilities, correlations change over time, especially in stressed markets. Ironically, when correlation is the most important—in stressed periods where you would really want diversification and therefore low correlation—correlations may increase. A common example is equity portfolios that contain domestic and foreign stocks. Suppose the average correlation between a domestic stock and foreign stock is 40%. The foreign stocks are there to diversify the portfolio with other economic regions. Suppose the markets are shocked, and the correlation increases because common global macroeconomic factors have affected both equity markets. Now, the average correlation is 70% instead of 40%. There lies the problem: the sense of security from the lower correlation and higher diversification is false because in a stressed market the correlation is higher and diversification is lower. As we've seen, lower diversification means the risks are greater. But even if markets aren't stressed, correlations—just like volatilities—can change over time. There's a model that allows you to look at volatility conditionally changing over time. This is a model from the same inventor of GARCH, Rob Engle, and the model for correlations is called the dynamic conditional correlation model. This is just to say that it could be treated as a variable.

6.9 Correlation affects other variables.

Let's revisit credit risk. Recall the formula for expected loss. The formula contains three variables. The expected loss is the product of the probability of default, the loss given default, and the exposure at default. Correlation seems like it doesn't come into the picture, but probability of default and loss given default are positively correlated. Get one wrong and you are likely to get the other one wrong too. In a relatively safe state of the world, there are low default probabilities and low losses given defaults. In a less safe state of the world, there are high default probabilities and relatively high recoveries. Estimate correctly and the expected loss is accurate. Underestimate and the loss is very understated.

6.10 There are alternatives to correlation.

The last property to mention about correlation is that sometimes you don't even want to use it. For example, in a non-stationary environment, you may prefer to find co-integrating variables that can be used instead of correlation. You may want to model joint distributions using copulas; in this case, a correlation value is still needed, but the bivariate relationship is modeled through the lens of a particular copula distribution.

Correlation is not the only way to describe how two variables behave together, but it is often used to discuss how things are related. No one on the financial news discusses cointegration, and even fewer people mention copulas. That's why we're in financial engineering: to enhance our toolkit. These are not difficult topics compared to other mathematics you will see. Once you've had some econometrics and stochastic models and machine learning, your toolkit equips you to eliminate assumptions and model more insightfully.

6.11 Correlation fails.

Correlation is one of the biggest problems in model risk because of the failure to understand joint dependency, particularly in stressed markets. Much of the regulation that came out of the global financial crisis helps to address flawed assumptions about correlation. Among other things, regulation has better prescriptions for large banks, things known as SIFIs, systemically important financial institutions, to perform rigorous stress tests that look at how their liquidity and solvency will stay intact given certain types of correlations that could happen. What would happen if the stock market crashed, liquidity dried up, and GDP tumbled? Stress tests are part of the annual compliance as regulation in many parts of the world now, and banks and other institutions that fail that are out of compliance and are not able to engage in transactions. Volatility can get very high, which can be seen in high VIX values, in data where we reach new highs and new lows, and in headlines about setting records for the most volatile trading day. But correlation is bounded; there is much less of a feel for how a small increase in correlation affects a portfolio.

7. Statistical Factors Combined: Structured Finance Tranches

In our tour of these financial challenges, we categorized the concepts in groups of two. They could occur in any combination. Let us take an example of how correlation and credit risk combine. To do so, let us examine the world of structured finance tranches. For simplicity, suppose you have a collateralized portfolio whose cash flows feed three different tranches:

- Senior tranche
- Junior tranche
- Mezzanine tranche.

The mezzanine tranche absorbs the first 15% loss of the portfolio.

The junior tranche absorbs the next 10% loss of the portfolio.

The senior tranche absorbs the final 5% of the portfolio.

Now, suppose there is LOW default correlation in the collateralized portfolio. Low default correlation means if one security defaults, the others are not affected in any direction. Zero correlation means uncorrelated. If the default probability is low, then we should have a distribution that is skewed. Consider now the situation of a stressed market. Now there is HIGH default correlation in the collateralized portfolio. Suppose it goes to a default correlation of one. Now, all the securities either survive or default.

Let us consider how the default correlation affects these two tranches. The equity tranche is adversely affected by low default correlation. A few defaults and the equity tranche could be wiped out. However, the equity tranche benefits from a high default correlation. Why? Suppose the default correlation is one. This puts the equity tranche in the same risk category as the senior tranche. Either everyone defaults and no one is paid, or no one defaults, in which case equity receives a higher return. The senior tranche benefits from a lower default correlation because it has the subordinate equity tranche top-subsidize the first set of portfolio losses. Thanks to low default correlation, it is highly unlikely that a massive number of securities default, so the senior tranche's payments are protected. However, the equity tranche suffers from a high default correlation. An extremely high default correlation concentrates the risk, putting all tranches into the same position: either all securities survive or they all default. In that case, the senior has the same risk but paid much more for the tranches because it was considered safer. Recall that senior tranches could be rated AAA, but equity tranches are not even rated.

Default correlation changes the risk-reward relationship in the structured finance world. Indeed, we could add volatility to this equation. How? Default correlation plays the role of volatility in the payoffs. This insight is one of the problems in your Group Work Project, where you'll see that the tranches can be thought of as option positions and where the underlying is the portfolio loss. As a hint, think of the type of option or options these positions have

MAGNIFYING AND FRICTIONAL FACTORS

Reading Time	70 minutes
Prior Knowledge	Volatility, Credit Risk, Financing, Volatility, Correlation, Basic pricing of stocks
Keywords	Difference types of leverage, Nonlinearity, Liquidity Characteristics, Regulation

In the previous lesson, we reviewed credit risk, financing, liquidity, volatility, and correlation. In this lesson, we'll review four more factors: leverage and non-linearity, regulation and capital adequacy. Let's begin with the magnifying factors of leverage and non-linearity.

1. Magnifying-Related Factors

Leverage and non-linearity are magnifying-related factors. They help to amplify returns. Leverage increases the absolute value of returns by simply having a greater size of investment. Non-linearity can amplify returns by having an influence that exceeds linear increments. There are different methods by which leverage and non-linearity operate. Let's begin with leverage and see different contexts in which it is used.

2. Magnifying-Related Factors: Leverage

Leverage can refer to different concepts or have different contexts. Let's visit each in turn.

2.1 Leverage as Force

Sometimes, people use the term leverage to describe a force or an exertion. Consider our example of a seesaw (or using the British term, a teeter-totter). You could have an adult and child stand on opposite sides of the pivot point of a seesaw. Yet, if the child sits further away, the extra distance provides a leverage that allows the child's force to equal that of the adult. How does this example apply to bonds?

Consider a bond's duration. Suppose you're faced with five cash flows for a two-year note (four coupon payments, one principal payment). Imagine you took a two-year note that has cash flows and a principal repayment. Now let's balance these along the seesaw. Bear in mind that the cash flows are proportional to their discounted values (so the discounted cash flows are not the same size). How does this example apply to a bond?

Consider a bond's **Macaulay duration**, or simply *duration*, (for a bond that pays coupons semiannually) assuming that the relevant time is $t = 0$:

$$MacD_{t=0,r} = -(1+r) \times \frac{P'(r)}{P(r)} = \sum_{t=1}^N \frac{CF_t}{(1+r/2)^t} \times \frac{t}{P(r)}$$

where, as before, r is the yield to maturity, $P(r)$ is present value of all cash flows (i.e., the current price (current as in $t = 0$) of the bond), CF_t is the cash flow at time t_t , with t_t being the time of t th cash flow.

This formula for duration was first proposed in 1938 by the Canadian economist Frederick Robertson Macaulay and perhaps the careful student might have noticed that we have (in the case of bonds that pay annual coupons):

$$MacD = (1 + r) \times MD.$$

The *duration* is the *weighted average time to receive the cash flows*. In fact, you should notice that the formula can also be written as

$$MacD(P)_{t=0} = \sum_{t=1}^N t \times w_t$$

with

$$w_t = \frac{CF_t}{P(r)(1+r/2)^t}.$$

Let's make a few important considerations.

Macaulay duration is always less than or equal to the maturity and, more precisely, it is equal to the maturity of the bond only when the bond is a zero-coupon bond.

Note also that if we increase the coupon size, we're adding heavier members in the front, which then shortens the duration. If we increase the maturity, we're adding distance to the principal payment, and this increases the duration.

Duration represents an equivalence: a zero-coupon bond that behaves like the original bond. Suppose you were to replace all these cash flows with a single zero-coupon bond with a maturity equal to the duration.

Then, we have two bonds that would experience similar sensitivities when the yield curve changes in small, parallel shifts. This risk equivalence was the original intention of the Macaulay duration.

Even though the units of duration are time, duration was intended to measure risk, as we stated in Lesson 1 of this module already, and it says that we can replace a particular bond with a zero-coupon bond whose maturity matches the bond's duration.

Of course, if the yield curve is shifting, if it's tilting as it is right now in many countries, we have an inverted yield curve. If it's going down, then that analogy doesn't hold, and that's why we can't just use duration to measure interest rate risk. Not even forget that all duration formulae presented so far require that interest rates along the yield curve move via parallel shifts or the formulae are inappropriate.

In this example, we're using duration as if it were a "lever," and this conveys the idea of leverage. To be clearer, suppose an investment manager expects interest rates to go down. We know that this would lead to an increase in the bond price. Then, the manager would want to lengthen duration; they extend the maturity to increase the leverage of the principal payment so it creates more leverage and moves the balance point out longer. On the other hand, if the manager expected interest rates to go up, then as this would lead to a decline in bond prices, the manager will try to manage this risk by lowering the duration.

2.2 Leverage as Financing

While we're discussing bonds, let's consider another way to think of leverage: borrowing money to increase the size of your investment. Most of the time when people think of leverage, they might think of borrowing cash. Suppose a trader has a strategy on which they make a ten percent return over a year. A \$100 investment would be worth \$110, earning a \$10 capital gain. This is just a straight cash investment. Imagine you can borrow \$50, combine that with \$50 of capital, and have that same \$100 invested. (For simplicity, let's assume 0% interest on borrowed funds). The \$100 investment is still worth \$110, earning a \$10 capital gain.

Now, when we compute the returns of each scenario, we notice that the first scenario returns 10%, while the second returns 20% (\$10 capital gain on a \$50 initial investment). The 10% return was doubled because the investment was leveraged 2-to-1. The two to one means you double the ten percent return you would have gotten, and that takes the ten percent return to twenty.

Recall the lesson in the module discussing leverage where you buy a home, say it costs \$200,000. Let's say that you put twenty percent down, and you borrowed eighty percent. In fractional terms, you're putting one fifth down (\$40,000) and financing 80% (\$160,000). Suppose the home appreciates ten percent to a value of \$220,000. You sell the home. You pay back the mortgage. Now you've made \$20,000 on an initial investment of \$40,000. That's a fifty percent return. That's because you were leveraged five to one. These results look very promising.

In fact, let's say that you're a hedge fund, and you're investing in some type of futures. Contract futures are among the most leveraged investments you can make. Suppose the fund invests \$5 down and borrows \$95. Suppose the investments makes 10%, or \$10. Calculating the return on the initial \$5 investment, we have a 200% return. Thanks to a leverage factor of 20-to-1.

The more leverage you have by borrowing funds, the more a return is amplified. Like volatility, leverage is agnostic to direction and works to improve gains or aggravate losses. Leverage is like time-lapse photography. Have you ever seen a nature show where the film is sped up so you can see the blooming of a flower over a period of seconds instead of weeks? That's what leverage does to returns; it speeds them up. Leverage helps small returns become large returns. The news is great for gains but terrible for losses. A leverage of 10-to-1 means a ten percent return becomes a 100% gain, but a 10% return means a 100% loss! Worse, a 20-to-1 leverage means a 10% drop becomes a 200% loss. Such losses could wipe out a trader, portfolio, or business! Where leverage is involved, risk management needs to provide checks and balances.

2.3 Leverage through Derivatives

Options do not allow financing: their premiums must be paid in full. Yet, options offer a second way to enact leverage: through the multiplier factor. For example, one equity call option grants the holder the right to buy 100 shares of the stock. Suppose you bought one option. Using boundary conditions, we've seen that this costs less than 100 shares of stock. Now suppose the stock price increases. This results in a much larger increase in the call price. There is not a simple number (e.g., 100-fold) due to the non-linearity and time value, but it is much larger than the amount the stock increases. Thus, derivatives provide leverage through a multiplier factor. Depending on the derivative, one contract tends to control many shares of the underlying securities.

This multiplier factor is true for futures. For example, one future contract could also be for 1,000 barrels of oil. Moreover, futures have financing built in as well. Many futures contracts typically require only a 2% to 5% deposit. So a bullish investor would be able to take an even larger position by entering a futures contract, which uses both financing and a factor multiplier to access a greater number of underlying securities by using less capital.

2.4 Leverage and Credit Risk

Everything that we've described previously is affected by leverage. Let's go through them one by one. As an example with credit risk, suppose you apply for a mortgage. One of the factors discussed was the debt to equity. Prior to the Great Financial Crisis (GFC), it was relatively easy to get mortgages where the homeowner puts down less than 20%. Say the home is \$100, and the down payment is \$20. The problem is if the value of housing drops more than 20%, the homeowner is effectively underwater. Recall the option analogy where the option is out of the money. We discussed this in Module 4. Prudent credit risk recognizes the leverage (debt-to-equity) of the borrower and ensures it can withstand temporary disruptions in income, housing corrections, or higher interest rates.

So if leverage is used and the result is a huge capital loss, there's a double whammy—the capital loss, and the need for payment of whatever funds were borrowed—the financing costs of using cash or other securities. Suppose the loss is enough to trigger a credit downgrade—after all, an extreme loss wipes out equity and increases debt. This downgrade could make the financing more expensive! For example, did you ever read the fine print on your credit card? If you miss a payment on a different card, and the credit card company finds out, they can raise the rate on their card. Why? They have learned the borrower is riskier, and the higher credit risk warrants the lender charging a higher rate.

Since the global financial crisis, many mortgage lending requirements have become stricter to prohibit debt-to-equity ratios that would be excessive. Fifteen years ago, the market was more competitive and relaxed about those; today, that has been made tighter. In part, these stricter controls helped prevent a housing crisis during the COVID-19 pandemic. If you look at the housing crisis during the GFC, there weren't the safety mechanisms in place that are in place today.

2.5 Leverage and Volatility

Equity exchanges have avoided the problem of excessive leverage by setting upper limits on the amount that can be financed. Now let's add in volatility. If volatility is 20%, leverage can turn that into multiples. Imagine in investment that is leveraged 20-to-1. A 1% return looks like a 20% return. But if volatility is sufficiently high to produce a short-term return of 10% then, with simple multiplication, the return is 200%! Extreme values that can make or break the portfolio. Using more volatility or using more leverage will get a portfolio to more extreme values than without. From the outside, it may be difficult to distinguish whether a return was made using more volatility ("risk") or more leverage. One difference is that leverage will always incur some type of financing costs, which still exist even in the case of huge capital losses.

2.6 Leverage and Correlation

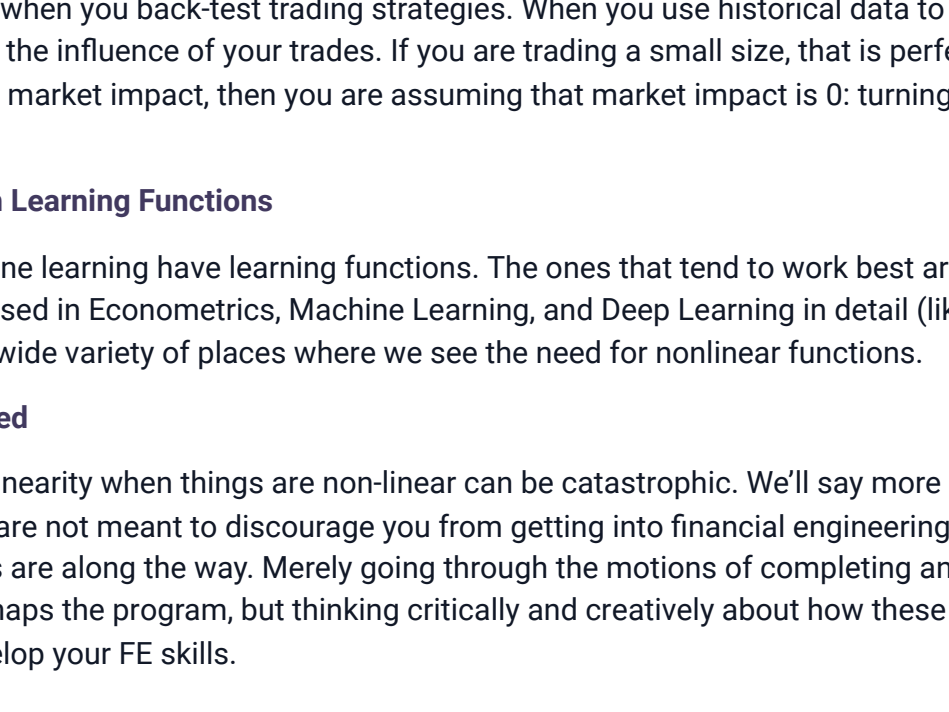
Correlation can amplify losses, particularly in credit risk. Consider the securitization examples you studied in previous modules. Higher default correlation causes the shape of the loss distribution to widen. Higher default correlation causes the securities in the pool to either survive together or default together. In a way, this leverages the returns to be extreme. Suppose default correlation increases to 1. Now we would have either complete default or no defaults. Indeed, here, default correlation is behaving like volatility. In your GWP3, think about the option position of the equity tranche.

3. Magnifying Related Factors: Non-Linearity

Oftentimes, we think of a system as linear. If we say that we invest four times the amount of capital, we should get four times the revenue. This illustration assumes we don't modify the amount of leverage. Sometimes, relationships between inputs and outputs are not linear. Let's discuss 4 examples of non-linearity.

3.1 Non-Linearity through Convexity

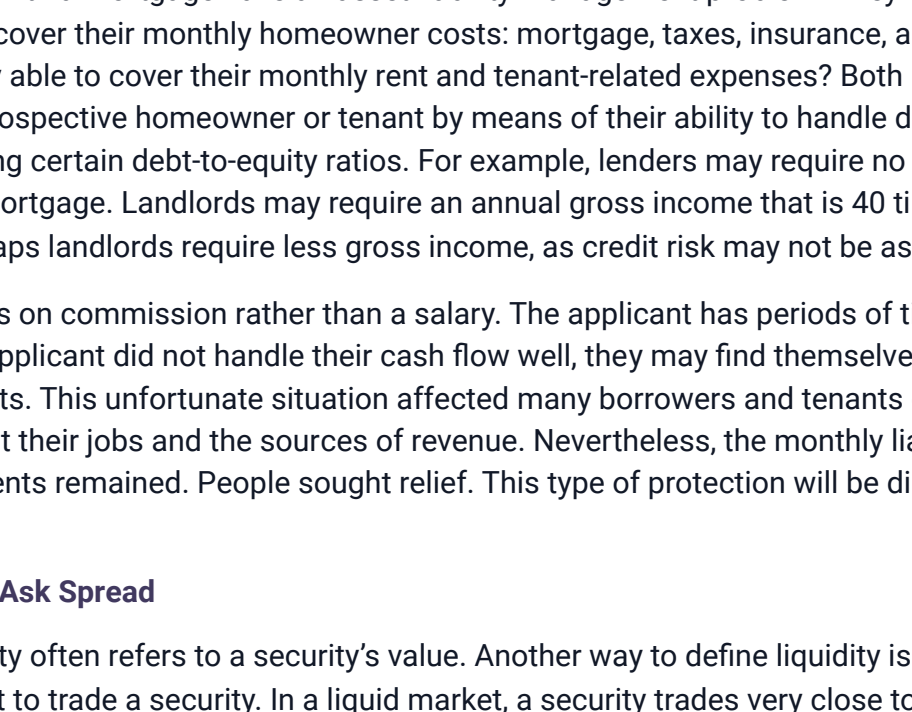
Let's graph a bond's price versus interest rates.



The relationship is both inverse and convex. Suppose we change the interest rates by 1bp. We get a price change called the Dollar Value of a Basis Point, known as the DV01. Suppose we change the interest rates by 10bp. Is the price difference now just 10 times the DV01? No, it is not due to the convex relationship. This is harder to deal with because when you're in the nonlinear world, this is an example that just has convexity. At Point A, yield changes produce relatively small price changes. At Point B, yield changes produce relatively larger price changes. Mathematically, convex relationships do not have a constant slope. In our bond price-interest rate graph, the slope changes from big negative values and then flattens as you increase the interest rate. The slope at point A is closer to 0 than the slope at point B. The non-linear world is harder, as we will see, because it is trickier to deal with maintaining the right ratio when hedging.

3.2 Non-Linearity through Options 1: Payoff

Let's graph an option's payoff at expiration.



The payoff is piecewise linear: flat when the option is out of the money (OTM) and sloped when the option is in the money (ITM). The strike is where the curve changes from flat to sloped and is known as the money (ATM). Although each portion is linear, the overall structure is non-linear. One way to think of non-linear is that the slope changes. At the strike, 100% of the change in slope occurs. Notice how the option payoff breaks symmetry. Putting a mirror on the strike would show that one side (OTM) is not a reflection of the other side (ITM). This means that call payoffs have unlimited upside but limited downside.

Suppose we are ITM. A small drop can keep the option ITM, but a large drop can make the option OTM. Therefore, changes in the underlying price stock do not necessarily result in proportional changes in the option payoff.

3.3 Non-Linearity through Options 2: Pricing

Recall that an option's payoff only applies at expiration. During the option's lifetime, an option has a price that we can model. Let's graph an option's price (using the Black-Scholes model you will learn later in the program). Note that the option's price is a nonlinear function of the underlying stock price. This is no longer piecewise linear but convex. Mathematically, notice how the curve is both smooth and continuous. This means the function will be differentiable.

3.4 Non-Linearity through Options 3: Sensitivities

The Greeks are the sensitivities of the option price to the underlying stock price, to interest rates, or the option hedge ratio or other products. All these sensitivities are non-linear. Recall that the slope of bond prices to time rates is also non-linear. Most products in fixed income and options have non-linear sensitivities.

One example of a security with a linear sensitivity is a stock. Financial practitioners often run the Capital Asset Pricing Model (CAPM), which produces a market sensitivity factor called beta. Beta is a constant. A stock's beta doesn't change very often. When you take the derivative of a constant, you get zero. There is no second derivative. In the world of equities rather than get more mathematical, you end up getting a larger picture. Equity traders tend to specialize in sectors, such as technology. They understand management, earnings, market share, profitability ratios, and so on. It's quite a different skillset than for fixed income, FX, or options. You'll learn that options have Greek sensitivities—deltas, gammas, thetas, and vegas—that are nonlinear functions of the underlying. The result of this property is that options often move more frequent hedging than stocks or bonds do.

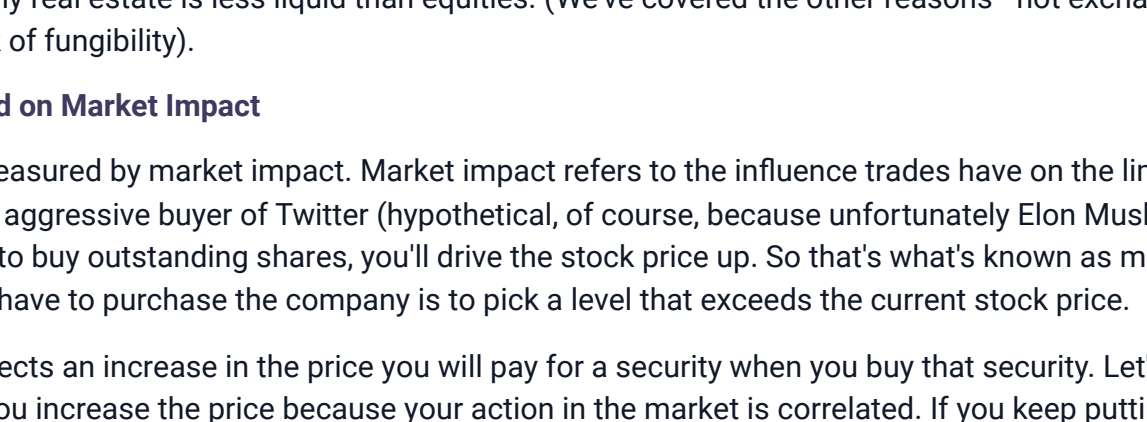
3.5 Non-Linearity through Variance

Recall from the module that discussed portfolio returns, the overall portfolio return is a linear combination of the individual returns, weighted by the proportion of each security's market value in the portfolio. Similarly, a portfolio of bonds would have a duration that is a weighted average of each individual bond's duration.

Variance is not linear. A portfolio of securities does NOT have a variance that equals the weighted average of each security's individual variance, except in the very special case where the correlations are equal to 1. The same is true for convexity. The same is true for option sensitivities. The disruption to this is correlation: in all but extreme cases (such as 1 or -1) we cannot simply use linear properties.

3.6 Non-Linearity through Liquidity

One final mention here is market impact. Consider a graph that illustrates market impact.



The x-axis shows slippage. Recall that slippage is defined as a price difference: the average price paid minus the price at the time the trade was entered. The x-axis shows trading volume. Trading volume is typically presented on a percent basis rather than outright basis. The graph conveys market impact. The greater the volume we buy, the greater the effect of market impact—typically, in a non-linear way. This illustration illustrates that a proportional increase in a volume traded could result in a disproportional slippage due to market impact.

The difficulty in detecting this type of non-linearity is that the effects can appear linear for small volumes. In other words, the effect may be proportional up to some threshold, and then becomes abruptly non-linear, much like an option payoff.

Market impact is a form of correlation. A large sell order influences buyers to lower bid, creating slippage, which is the measure of market impact. Conversely, a large buy order can cause sellers to raise their ask prices, triggering market impact. Keep this in mind when you back-test trading strategies. When you use historical data to back-test a trading strategy, it did not contain the influence of your trades. If you are trading a small size, that is perfectly fine. If you are trading a size that creates market impact, then you are assuming that market impact is 0, thus creating a non-linear effect into a constant of 0.

3.7 Non-Linearity through Learning Functions

Many algorithms in machine learning have learning functions. The ones that tend to work best are the nonlinear ones. Of course, this will be addressed in Econometrics, Machine Learning, and Deep Learning in detail (like ReLU and Softmax). For now, we're just listing the wide variety of places where we see the need for nonlinear functions.

3.8 Non-Linearity Reviewed

The danger of challenging reality when things are non-linear can be catastrophic. We'll say more about this in model failure. These assumptions are not meant to discourage you from getting into financial engineering. They're really meant to show you what the pitfalls are along the way. Merely going through the motions of completing an assignment can get you through a course and perhaps the program, but thinking critically and creatively about how these models are addressing these challenges will develop your FE skills.

4. Frictional Risk Factors: Liquidity and Regulation

Liquidity and regulation are to financial markets what air resistance and friction are to physics. When you study physics, you initially ignore air-resistance and friction to make problems tractable, easier to solve, and simpler to interpret. Nevertheless, these frictional resistances exist in the real world in the form of regulation and liquidity. In more advanced physics or engineering, they are accounted for when modeling physical systems. Likewise, liquidity and regulation are costs we must anticipate and calculate within models. Many books discuss the topics interchangeably, for example (Darrell Duffie's "Fragmenting Markets: Post-Crisis Bank Regulation and Financial Market Liquidity"). Let's begin with liquidity.

5. Liquidity

Let's consider different contexts in which liquidity is used in financial markets.

5.1 Liquidity as Solvency

At the individual or institutional level, one could assess liquidity by asking: Are there sufficient cash flows to pay the debt/bills? Is there enough cash flow to handle unexpected expenses that arise? These are questions of solvency. These questions pertain to individuals or companies that are faced with making debt payments over time.

For example, homeowners with a mortgage have an asset liability management problem. They have monthly income and expenses. Are they able to cover their monthly homeowner costs: mortgage, taxes, insurance, and utilities? Lenders have the same problem. Are they able to cover their monthly rent and tenant-related expenses? Both mortgage lenders and landlords will assess the prospective homeowner or tenant by means of their ability to handle debt. Lenders and landlords assess this through reviewing certain debt-to-equity ratios. For example, lenders may require no more than 40% of the income going toward the mortgage. Landlords may require an annual gross income that is 40 times the monthly rent. (In more lenient markets, perhaps landlords require less gross income, as credit risk may not be as much of an issue).

Suppose an applicant works on commission rather than a salary. The applicant has periods of time through the year where income decreases. If that applicant did not handle their cash flow well, they may find themselves behind on mortgage payments or rental payments. This unfortunate situation affected many borrowers and tenants during the COVID-19 pandemic. Many people lost their jobs and the sources of revenue. Nevertheless, the monthly liabilities of mortgage payments and rental payments remained. (NBC). Within financial firms, all communication today is likely recorded: emails, texts, chats, posts, etc. Brokers who lie to customers are not performing their fiduciary responsibility. In the most egregious cases, some of the brokers went to prison (you'll read more about this in the next lesson).

5.2 Liquidity Based on Bid-Ask Spread

Aside from solvency, liquidity often refers to a security's price. Another way to define liquidity is how far from the true market value we can expect to trade a security. In a liquid market, a security trades very close to its true value. In an illiquid market, a security trades away from its true value.

The bid-ask spread is one of the most common ways to measure liquidity. From the sell-side (market maker) perspective, the bid-ask spread represents the profit margin the sell side extracts for providing a two-sided market. A market maker is willing to sell at the ask price and buy at the bid price. A wide bid-ask spread means that the market maker is willing to sell at a high price (ask price) or buy back the security at a low price (bid price). A tight bid-ask spread means that those 2 prices are closer to each other. Typically, the mid-quote is the average of the bid and ask price.

The bid-ask spread depends both on the security and on the type of market—exchange, OTC, retail—where the transaction occurs. For example, suppose you arrive in Tokyo, Japan, and want to convert your currency (say one million Nigerian naira) to Japanese yen. You're purchasing yen at the ask price, then, to discover you must return home immediately. Now you're selling yen at the bid price. The transactions happened within such a short time, there was no change in prices. You'll wind up with fewer naira than when you started because of the bid-ask spread. Instead of transacting at the airport, suppose you transacted with your global bank when using their ATM. The exchange rates would likely be much better, meaning that the bid-ask spread was tighter. Therefore, the same FX transaction, naira to yen, would be more liquid electronically through a bank than through the airport kiosk. The kiosk has significant overhead (rent, labor) and a much more limited set of transactions than the bank and uses the bid-ask spread to increase the revenue to offset those relatively high expenses.

One huge advantage of exchanges is their ability to show market participants the limit order book. Suppose you are an aggressive trader and want to buy shares of stock. You could get the full ask size at the current ask price and second best ask size at the second best ask price, and so on. You could get as much as whatever the ask size is at that time. On the other hand you, suppose you place a limit order. Suppose you're only willing to pay a maximum of \$9. Instead of taking supply from the market, you're adding to it as the supply of the market. The limit order book will enter your price and size (in the order in which the trade is received) until it is filled or canceled. Instead of taking liquidity from the market, you're willing to wait until someone is willing to sell to you at \$9. There is a chance you do not get filled. Suppose the price only dropped to \$9.05. Then, you would have protected your limit of \$9, but you did not get it filled. The lost opportunity may not have been worth it, if you feel the stock was undervalued 20%. The market order is different from a limit order because the market order gets filled, but perhaps at an unattractive price, whereas the limit order may not get filled, which is execution risk.

The bid-ask spread can also vary according to the time of day. Suppose you trade NFLX stock during active NYSE hours. The bid-ask spread will be relatively tight. Suppose you trade 12 hours later, when the NYSE is officially closed. There are only a few market makers providing liquidity overnight—but they do so at a cost that widens the bid-ask spread considerably. So the bid-ask spread depends on many factors: the security, the location, the time of day.

On the day before a holiday weekend begins, many market makers head out on vacation early. Not surprisingly, there are fewer skilled market makers. At these times, one can find the bid-ask spreads larger as this period behaves like an off-hours market.

5.3 Liquidity Based on Volume Traded

Another way to assess liquidity is to assess the volume that trades. On the crypto exchange called Binance, Bitcoin trades an enormous amount of volume. Typically, \$11 to \$12 billion of average daily volume is common. The S&P 500 is a common equity index containing the top 500 stocks by market value. Interestingly, the top 50 stocks (top 10%) on the index make much higher average daily volumes than the bottom 50 stocks (bottom 10%). So the top stocks are more liquid via their volume. The active (on-the-run) U.S. Treasury benchmark securities have significantly more average volume than that of the off-the-run securities. Moving to corporate bonds, some bonds may not trade for days or weeks. Recall that some portfolio managers hold bonds until maturity, redeeming their coupons along the way. Effectively, this means those particular bonds are not very liquid as measured by the volume. Finally, options can have both a volume traded and new issuance. Recall that options are bilateral contracts, and so new ones can constantly be written as demand grows. Recent studies showed that some of the Bitcoin trading volume may be misleading (Paz).

5.4 Liquidity Based on Profitability

Greater volumes can be traded when the items being traded are identical to each other. One share of Netflix common stock is like any other. Japanese yen are indistinguishable from each other. Five Bitcoins in your account are equivalent to 5 Bitcoins in another person's account. One hundred barrels of crude oil (with certain specifications) are like any other 100 barrels of crude oil.

Yet, the same may not be true for diamonds, which vary with several features (cut, color, clarity, and carat). The same is not true for homes because no two homes are exactly alike. Yes, they will have the same floor plan, materials, etc., but they cannot physically be in exactly the same place.

Fungible securities (stocks, dollar bills, Bitcoin) are interchangeable. Non-fungible securities (diamonds, homes) are not interchangeable. An extra level of due diligence is required at the individual security level when trading non-fungible securities. For example, it may be a buyer's market in real estate, but one still has to perform prudent valuation of a specific property. Generally speaking, markets with fungible products are more liquid than markets containing non-fungible securities. Examining the idiosyncrasies of non-fungible assets takes time. Homes take longer to examine, finance, and close than stocks. You could buy and sell stocks several times a day. If you're a high frequency trader, you could do that thousands of times a day. It would take weeks or months to flip residential or commercial real estate because of the paperwork and complexities involved.

5.5 Liquidity Based on Time to Trade

Liquidity is also measured by the time it takes to get a trade executed. During active NYSE hours, trading stocks is quick. The same is true for stocks on NASDAQ. Now suppose you want to trade real estate. You could trade Real Estate Investment Trusts (REITs) as one of those exchanges very quickly. However, if you were to purchase a residential property, that would take weeks. The process is time-consuming, and depends on various entities (mortgage lenders, seller, realtors, appraisers, inspectors, local government officials) and is not intended to occur within the same day. This process is one of several reasons why real estate is less liquid than equities. (We've covered the other reasons—no exchange traded, not as much volume, lack of fungibility).

5.6 Liquidity Based on Market Impact

Liquidity is also measured by market impact. Market impact refers to the influence trades have on the limit order book. Suppose you're an aggressive buyer of Twitter (hypothetical, of course, because unfortunately Elon Musk beat you to it). As you put bids in to buy outstanding shares, you'll drive the stock price up. So that's what's known as market impact. The only chance you'd have to purchase the company is to pick a level that exceeds the current stock price.

Market impact reflects an increase in the price you will pay for a security when you buy that security. Let's say you want to start buying, you increase the price because your action in the market is correlated. If you keep putting bids in to buy, then your action is going to increase the price, simply because other market participants, the market makers who are selling will be motivated to increase the value, thinking that you'll be willing to pay the higher price. There's a price increase as well as a time that it needs to bounce back. The market impact is like the rebound of a spring. A liquid security has a spring that reverts quickly. Then, you're stretching the spring a great deal, so you increase the price quite a bit from the impact, and then it's also slow to return to its equilibrium position. If something is very liquid, which means it has little market impact, there will not be a big price increase, and it will take a very short amount of time to return to its original value.

Recall from the nonlinear section the discussion of market impact. A profitable trading strategy can turn into an unprofitable one. The expected alpha of the trade may be eroded by the market impact faced in each direction: facing slippage on entry by paying more than the current ask price and facing slippage on exit by selling for less than the current bid price. To avoid becoming predictable in your strategy, execute algorithms will slice and dice large orders to avoid detection from algorithms of the intent of the large buy order.

5.7 Liquidity Affects Credit Risk

When it is time to pay a debt, is there enough cash flow to pay the liabilities? This question is the core problem of asset liability management. From a solvency perspective, liquidity is important to assess credit risk. Imagine that a company has no debt, no cash flow requirements (salaries, rent, expenses, etc.). In this unusual case, liquidity is not an issue because there are no impending liabilities. Indeed, many growing companies are in a phase where they have pools of capital (venture capital, private equity, angel investors), are not issuing debt, and may not be expected to be profitable during a ramp-up phase. It often takes years to build a factory, patent a drug, or start a brand that becomes a profitable company. Expenses along the way create possibilities for the firm to default. Oftentimes, these expenses or negative profit margins create a burn rate that requires startups receive additional funding to keep going. The real test of liquidity is really to face liabilities, and the inability to do so is certainly a form of credit risk.

5.8 Liquidity Review

In short, measures of bid-ask spread, volume, time to trade, and market impact are all considered when executing trades. Bid-ask spread is the most popular because it is easier to observe, and acts like a cost (e.g., 20 basis points). Suppose you buy a stock on an exchange at the ask price. The prices don't change. Then, minutes later, you decide to sell the stock. You sell at the bid price. Bouncing up, you paid one bid-ask spread. Some students think you pay it twice, but remember that each trade is paying about half the spread (ask price minus bid price going in, and bid price minus bid price going out). In some markets, the granularity between prices is quite small—one cent in stocks. In other markets, the intervals can be much further apart (e.g., \$32.50 in Treasury markets).

5.9 Liquidity as Solvency Revisited: The Banking Level

We discussed liquidity risk at the individual or company level: will an entity have enough funds to pay its debt? Just like corporations, banks should have sufficient collateral or cash reserves to weather bad storms. While a bankrupt corporation is unfortunate as it causes job losses and regional economic decline, a bank declaring bankruptcy is worse. It erodes trust in the financial system, and possibly affects all the banks in the industry—both those with which it engaged in transactions (i.e., interest rate swaps) and even banks with which it did not because of the interconnected nature of the global financial industry. This notion of enforcing bank liquidity brings us to our next challenge: regulation.

6. Regulation

Regulation exists for many reasons: people or companies over-speculate, behave unethically, and behave unintentionally. The need to produce revenue may cause an excessive exposure of risk. Often, leverage and non-linearity contribute to this tremendously. Anyone who is OTM—a trader or portfolio manager who is down on the year, a CEO of a company that will have no income for a quarter—is incentivized to take leveraged risks. Big wins will put them deep ITM, with continued employment, big bonuses, and stock option valuations. Big losses, on the other hand, may not put them OTM, but their firm. If a trader loses \$X for the year, the firm can still survive, but the trader may lose their job. If a trader leverages to try to make \$9X or lose \$10X, they are no worse off for the loss but have a potentially big gain. Unfortunately, the trader's incentives are not aligned with those of the firm.

Let's divide our study of regulation into several groups. Let's start with regulation at the asset level.

6.1 Regulation at the Asset Level

Consider a firm that wants to exit a portfolio of mortgage-backed securities. As you recall, MBSs do not trade on exchanges but OTC. Whereas exchanges would have publicly transparent pricing, OTC is less transparent. Instead of prices, there may be indications. There are likely to be delays as well.

The regulation known as TRACE (trade reporting and compliance engine) requires that they report the trades within 15 minutes. TRACE reporting applies to Treasury bonds, agency bonds, corporate bonds, MBSs, and other asset classes. Firms must invest

FINANCIAL ETHICS

Reading Time	45 minutes
Prior Knowledge	Market Risk, Credit Risk, Market Making
Keywords	Fiduciary Responsibility, Spoofing

1. Introduction to Financial Ethics

For Adam Smith, “a free market must reset on a sound ethical foundation” (The Ethics Centre). Ethics matter. Financial markets exist to the extent that people behave ethically. Recall from Module 1, if a businessman fell behind on his debt, his creditors would break the bench on which they conducted business, giving us the Italian word for “broken bench”: bankrupt. Back then, the industry regulated itself. The scope, complexity, and volume of finance today would bewilder businesspeople from thousands of years ago. Ethical dilemmas exist in areas today that didn’t exist even a generation ago. In this final lesson of the module, we will look at ethics in different categories.

1. Taking Things That Aren’t Yours: This includes plagiarizing and taking code that doesn’t belong to you.
2. Hiding Things That You Shouldn’t Hide: This relates to a lack of internal controls and leads to hiding trading losses that can jeopardize the health or existence of a financial institution.
3. Lying to Colleagues or Customers: This includes providing customers with false prices, fabricating emails that relate to hiding trading losses, or failing to disclose the risks of an investment.
4. Trades That You Shouldn’t Do: These examples include insider trading or spoofing markets.
5. Failure to Disclose Risks: This includes neglecting to act in the client’s best interest by failing to disclose risk.

2. Taking Things That Aren’t Yours

It is important to act ethically when “no one is looking.” If you knew doing something was wrong, but believed you wouldn’t get caught, would you still do it? This is an example of an ethical question. One need not be working in the financial industry to face this temptation. This temptation arises each time a student is faced with what seems like a daunting assignment as the student ponders resorting to plagiarism. Our recommendation—in fact, our school’s policy—is not to do it. Of course, most schools say the same; some, including WorldQuant University (WQU), even go to the point of having students sign an honor code asserting that they will maintain the integrity of the institution with statements like “Do your own work” and “Give credit where credit is due.”

The “*taking things that are not yours*” issue can present itself even at work. When you work for a company and create software to solve certain problems, unless the company explicitly tells you otherwise in a contract or some kind of agreement, the software you created belongs to the company that employs you. That means that if, for any reason, you no longer worked for that company, you cannot bring that code with you in any form and use it. In particular, the worst possible thing would be to use that code you developed when working for company X to the advantage of company Y. In this case legal actions could be brought against you in certain cases. However, you could not very likely be forbidden to recreate the software line by line from scratch. And this is another reason why simply being able to use code other people developed does not make for valid quantitative analysts since as soon as they lose access to a certain program or are unable to steal it, they are basically unable to perform the tasks asked of them.

Same goes for lists of clients. Those clients belong to the company even if you were the ones contacting them on behalf of the company. Again, evading this obligation could also cause your previous employer to bring legal actions against you.

But let’s consider the academic issue of plagiarism. It is possible to outline three different forms of plagiarism that, sadly have become relatively common due to the ease with which people have access to internet.

1. Using appropriate references without citing them.
2. Using inappropriate references—that is, blatantly copying large portions of a previous submission from a previous group or semester.
3. Uploading your own solutions to publicly available course portals.

Regardless of the type, all plagiarism is wrong. Let’s take a dive into how plagiarizing equates to stealing, by examining the 10 groups of people that are the victims of this theft.

First and foremost, plagiarists steal from the original source—they try to pass off other people’s work as their own. The researcher, author, journalist, or blogger who performed their own hard work, critical thinking, and creativity to produce a journal article, book, or blog post is denied credit. Using that source without citing it is the most classical form of plagiarism. Our recommendation is two-fold: first, be sure to cite it properly within the text and within the references. Second, be sure to add original content that connects the work to the analysis of the particular argument or assignment. If you do this, your work will look more purposeful in that you are deliberately utilizing it to support your own point of view or perspective on the issue at hand.

Second, plagiarists steal from their group mates—specifically, they take away the opportunity for their group mates to earn a passing grade on a group work project. Remember that at WQU, plagiarized assignments earn a zero, and each individual in the group earns a zero, not just the plagiarist. In this program, students collaborate in the form of a group work project more than two dozen times. One of the most important parts of this educational experience is building teamwork, camaraderie, and trust with one’s peers. By plagiarizing, the culprit violates the trust of their peers, and significantly diminishes the opportunity to build connections and engage in fruitful collaboration. Frequently, the group disbands after that happens, and each person’s learning experience is weakened. Some plagiarists believe that just the plagiarized portion earns a zero, or perhaps just the person who plagiarized. This is 100% **not** the case. Plagiarism is a poison, tainting the entire submission, not just the portions that have been plagiarized. The easiest way to build animosity and resentment is to plagiarize and damage the grades of one’s innocent groupmates.

Third, plagiarists steal from their classmates—specifically, they steal their time and effort by taking a shortcut that gets them to the end goal: a completed project. Suppose a plagiarist spends very little time and effort on a paper, doesn’t get caught (a probability to be discussed later), and somehow earns an inflated grade. This is unfair to students who earn their grades through persistence and hard work. Moreover, the plagiarist may be emboldened to brag to students how easy it was, influencing other students who observe the “easy way” and may be tempted to plagiarize as well. Some students who did not do as well may simply question why they spent so much time and effort and still achieved lesser results. This act sets a bad example by claiming results without work.

Fourth, plagiarists steal from the graduates in the program—specifically, they adversely affect the reputation of the program, diminishing the value of a WQU degree in the marketplace. Again, suppose a plagiarist somehow managed to get through the Risk Management class and earn a good grade. Imagine that person goes on an interview and gets questioned about certain topics but is unable to provide informed or competent answers. This is plausible because the student never had to do the hard work to learn the concepts during their studies. The plagiarist may end up embarrassed by the interview, but employers may quickly begin to associate the program with inflated grades and underqualified students. Consequently, employers will become discouraged from interviewing or hiring other graduates from the program. This theft is serious because it creates a real opportunity cost for students who worked hard and followed the rules.

Fifth, plagiarists hurt future students. How? When they upload course notes to a website, plagiarists create a temptation for those future students to plagiarize for themselves. University policy forbids this, but the real damage again is to students who follow the rules. Stored files weaken the reputation of the overall program. Uploading solutions tempts students in the future to use previously completed work rather than build their own skills. Just because a file is uploaded doesn’t even mean that it is correct! Never upload course content or completed assignments to a portal.

Sixth, plagiarists steal from their professors—specifically, their time and efforts. Teaching faculty truly want students to learn the material the right way—otherwise, they wouldn’t be teaching. Faculty are happiest when students learn and grow. Grades are not meant to be a prize but a reflection of a student’s dedication and performance. When a student plagiarizes, they attempt to replace that dedication with someone else’s work passed off as their own. It undermines the efforts of content developers, reviewers, and editors; lecturers; graders; and other faculty members who provide materials, share experience, and offer feedback for the growth and development of students.

Seventh, plagiarists steal from the administrators of the school. The mission of the university is “to make advanced, quality education accessible to capable students everywhere by leveraging technology at scale to deliver entirely free online programs.” It does so at no cost to the student. The existence of this school is a testament to how the founders and administrators want to make an accredited master’s degree in financial engineering widely available to all by removing hindrances due to cost. While the school does not require money, it does require integrity. It makes no sense to steal from the institution that has invested resources and personnel in you and your professional development by diminishing the school’s brand in the marketplace. Plagiarizing is an affront to the university by threatening its ability to fulfill its mission. This is the reason why plagiarists are dealt with harshly. Please respect the mission of the university.

Eighth, plagiarists steal from their future employers—specifically, they are using deception to take a position for which they may not be qualified. A future employer assumes that the student has completed the rigorous financial engineering workload required for a two-year master’s program. This comprehensive qualification is one that most people in the industry don’t achieve. Even if the plagiarizing employee continues to do good work for their employer, their work is predicated on a foundation of inflated, or even fabricated, skills and competencies. People often get fired if they were found out to have plagiarized.

Ninth, plagiarists steal from their future coworkers. By presenting an embellished skill set on a team, plagiarists pretend to be something they are not. Coworkers will expect the employee in question to have the required skills yet may notice that person engaging in deflection, postponement, delegation, or any other method to avoid doing work they are supposed to know. The longer the plagiarist is an employee, the more difficult it is to come clean and admit they don’t know something. This may even embolden the plagiarist to continue lying or stealing. This creates the problem of false pretenses. Here’s a well-publicized example. The Dean of Admissions at MIT, one of the most prestigious technical schools in the world, lied on her resume (Martin). After a 28-year career at the school, the lie was uncovered and the Dean was fired, wiping away decades of her good service, which was all built on false pretenses. Reputations are like credit risk: it is difficult, if not impossible, to bounce back from a “default” state.

Finally, plagiarists steal from themselves. Consider a student starting out in the program. They will dedicate some 1,500 to 2,000 hours to get a master’s degree. Once they earn a master’s degree, they have it for life! (Some certifications must be renewed periodically.) This will become part of their professional profile. If they think about how relatively small any one assignment or project is in the scope of the entire master’s program, they will not be tempted to plagiarize. The student realizes that if they do plagiarize, potentially all their effort could be for naught. Why? Because a single violation of the Academic Integrity policy due to plagiarism earns its culprit a failing grade for the class or termination from the program. Is one assignment worth it?

Plagiarism in this day and age extends beyond copying from some external source. The introduction of ChatGPT and similar AI tools has made it possible to avoid doing the work by simply asking the tool the same questions that were asked of students and then copying and pasting the answers. Although the use of AI tools is not forbidden while a student is learning certain materials in a class, it is equivalent to cheating if those tools are utilized during tests (such as graded quizzes, group work projects, etc.). WQU takes this very seriously, and heavy penalties are imposed when AI tools are used.

We’ll end our discussion of plagiarism with a comparison to credit risk modeled via binomial trees. Suppose a business has to pay back a debt. There are two outcomes: they can either pay it or default, as illustrated by this binomial tree.

Business owes debt. Either the business pays or defaults.

Default is an absorbing state. This means that once the business defaults, it stays in a default state. They would declare bankruptcy and be forced to restructure or go out of business.

Now consider a student who has an assignment due and is considering plagiarism. There are two outcomes: they can either do their own work, or they can plagiarize, as illustrated by this binomial tree.

The student owes an assignment. There are two paths:

- *Does the student do their own work? This leads to a grade based on quality of work.*
- Does the student plagiarize? This leads to different grades.
 - Grade of stolen material (not necessarily an A)
 - 0 if caught—*not only for the cheater, but for their group as well.*

If students do their own work, they learn and grow and receive a grade for their results. If the students plagiarize, they bypass the learning and growing phase and receive a failing grade for the course.

What do these two problems have in common? The probabilities are *unknown*. One of the difficulties in credit risk is that default is relatively rare. In this program, we will spend time studying how to calculate the default probabilities by looking at stocks, corporate bonds, credit default swaps, and credit ratings. Similarly, one of the difficulties in *deciding* to plagiarize is estimating the probability that you will get caught. Students may not estimate at all, thinking “they will never get caught.” Ironically, this lesson is one that will be taught in Behavioral Finance, underestimating probabilities when it applies to them personally.

One thing to know is that the university is equipped with cutting edge tools for detecting plagiarism. For example, Turnitin is incredibly effective at finding plagiarized content. Indeed, students must run their assignments through Turnitin before submitting them. The list of sources that Turnitin checks is constantly expanding. Therefore, the potential plagiarist may think they know the probability of getting caught, but it is likely that they underestimate this.

Moreover, the student who plagiarizes tends to *underestimate the loss* given they are caught. When they inevitably get caught, every plagiarist is sorry after the fact. The excuses tend to include some combination of “I’m sorry,” “I didn’t know,” and “I won’t do it again.” While the apology may seem or even be sincere, the plagiarist wanted a good grade. This phenomenon is another variable in credit risk known as the LGD: Loss Given Default or the recovery rate (1 minus the LGD). When the revealed plagiarist negotiates for just a few points off rather than an official 0, they are attempting to set their recovery rates. For example, they plead, “Can you just take the 10 points off on the question I plagiarized instead of the whole assignment?” As we previously addressed, plagiarism is poison that taints the whole assignment. The plagiarist should expect a 0 on their assignment, which amounts to a recovery rate of 0. Furthermore, many faculty will not write letters of recommendation or work with students on their Capstone project if the student has a history of plagiarism. One’s reputation is one’s brand. Our advice is to maintain the brand’s quality by working professionally.

There’s even another variable in credit risk that relates to plagiarism: default correlation. Someone who plagiarized one time is likely to have done it before—and to do it again. In other words, the plagiarist’s behavior is likely to be correlated across assignments and classes.

Maintaining good financial ethics is one of the program outcomes. As a student, avoiding plagiarism is one of the best ways to build these ethical practices. Now, let’s look at other examples in industry where ethics can be an issue.

One other example of “taking things that aren’t yours” comes from industry. Sergey Ilyenikov stole high-frequency trading software from his employer Goldman Sachs. Then, he quit and accepted an offer at a hedge fund. Stealing the code triggered security alerts, which notified his employer of his theft. He was caught, tried, convicted, and served time in jail (Henning). There is much more to this story, which became the basis of the best-selling book *Flash Boys* by Michael Lewis.

Yihao Ben Pu did the same thing: He stole code from his employer and attempted to bring it to another hedge fund. He was also caught, tried, convicted, and served time in jail (USAO).

Stealing code from one’s employer may not seem like a crime, especially if that person wrote the code themselves. But a careful reading of an employment contract will show that the code belongs to the company rather than the employee. In fact, this code can be one of the most valuable assets a hedge fund has if it describes trading strategies. Our advice is to not take what is not yours, whether it is someone else’s assignments or code.

3. Hiding Things That You Shouldn’t Hide

A second set of ethics addresses “hiding things that you shouldn’t hide.” Our first example is a classic one because it occurred nearly 30 years ago and is often the first case study in risk management for financial crises. It was also turned into a Hollywood movie (see the 1999 movie *Rogue Trader* starring Ewan McGregor). Barings Bank employee Nick Leeson made inappropriate derivative trades and hid the losses, to the point that the catastrophic size was enough to bring down the 230-year-old financial institution (Smith). Sometimes, the losses are hidden not by a single individual, but by multiple people in an organization. For example, Daiwa Bank admitted to hiding more than \$1 billion in trading losses (Truett). This loss was significant but did not cause Daiwa to go bankrupt. People at the firm pled guilty, causing the firm to pay a record fine of \$340 million.

Risk management was in place at all these institutions, but a combination of unethical behavior and negligence contributed to these becoming epic financial disasters. This is not merely market risk—it is the lack of controls, checks, and balances to protect the firm from its own unscrupulous employees.

4. Lying to Colleagues or Customers

A third set of ethics addressed lying to colleagues or customers is wrong. Lying to colleagues may not seem so bad, but what if those lies were to hide losses? Jerome Kerviel at Société Générale lost \$7.2 billion in trading losses. This was 1.5 times the market capitalization of the bank itself! He had hacked the company’s surveillance system to bypass the proper risk channels. He traded enormous sizes without taking the corresponding hedges. As he lost money, he fabricated emails that made it look like his positions were being monitored and managed appropriately. Eventually, he was caught, tried, convicted, and sentenced to prison. However, he was also fined for the amount he lost: some \$6.7 billion dollars. Clearly, he doesn’t have that kind of money, but he remains in debt to the bank for that amount (Ayandayo).

When it comes to customers, there is much more formal regulation in place. Oftentimes, traders, asset managers, brokers, or dealers must act as a **fiduciary**, which means the agent acts on behalf of a client to serve the client’s best interests. Sometimes, advisors lie to their clients because they have huge losses on their customer’s accounts, but don’t report these directly to the client. This could take the form of a Ponzi scheme (as in the Madoff example), simply losing money and not telling anyone (for one example, see Longo), or simply using client funds to subsidize a lavish lifestyle until one is caught. If you google the words “advisor,” “using client funds,” and “lavish lifestyle” (without quotes), you will find thousands of articles of offenders. This relationship is built on trust. Breaking that trust violates the fiduciary responsibility of the fiduciary.

5. Failure to Disclose Risks

Another group of ethics violations occurs when financial professionals fail to disclose risks. Ethics can break down when a trader, dealer, or advisor fails to work in the best interest of the customer. These are well explained in the interesting article by Claire Hill: “Bankers Behaving Badly.”

This article addressed different types of cases. The first type of case is where advisors sell unsuitable investments to clients. A well-known example involves former Goldman Sachs banker Fabrice Tourre, who sold complex securities to customers who were not suitable for these products, failed to disclose the proper risks, and did not work on behalf of clients (U.S. Securities and Exchange Commission, “Goldman Sachs to Pay”). Another type of case is where advisors and clients both work unethically to deceive or defraud a third party. The Morgan state swaps disaster in Jefferson County is a classic example (U.S. Securities and Exchange Commission, “J.P. Morgan Settles”). Representatives of a Wall Street firm were charged with making unlawful payment to county commissioners to agree to a deal that benefited the bank and the commissioners personally. However, the transaction led to the bankruptcy of a county—one of the largest municipal bankruptcies in the entire United States. The bankers did not disclose the risks and paid off the commissioners, who accepted the higher than usual rates on the swap payments. Think of this as a two-stage fiduciary responsibility, in that both the bankers and the commissioners did not act in the best interest of their constituents, respectively. Fines were paid and jail sentences were served. These types of swap disasters are unfortunately all too common, and a book was written about them (see Abdel-Khalik’s *Brazen: Big Banks, Swap Mania, and the Fallout*).

Other times, failure to disclose risks can help cause a firm to go under. See the Jeffers article on how failing to disclose the risks of repo transactions caused Lehman Brothers, a 100+ year old company, to go bankrupt during the Global Financial Crisis. Thanks to regulation that arose out of the Enron Crisis (Sarbanes-Oxley), measures were in place to show how the accounting was fraudulent.

These types of unethical salesmen helped motivate the expression “caveat emptor,” let the buyer beware. Prudent financial professionals should know the risks of the transactions they enter on behalf of their constituents, clients, and companies.

6. Trades You Shouldn’t Do

In this last section, let’s look at “trades that you shouldn’t do.” The first is insider trading. Insider trading means trading on information that is not materially public. Mathew Martoma was sentenced to nine years in prison for insider trading that involved approximately \$275 million in illegal profits and avoided losses. Interestingly, this was not the only dishonesty in his history. He had been enrolled in Harvard Law School but was expelled for grade manipulation and dishonesty. Later, he attended the Stanford Graduate School of Business and earned an MBA. However, when Stanford learned that he had failed to disclose his Harvard expulsion, Stanford revoked his MBA. “False pretenses” has no expiration date, as we saw earlier for the MIT Dean of Admissions. This means that the threat of being exposed for plagiarism or other lies can ruin your reputation and work that was done earnestly and honestly.

An even more famous case than Mathew Martoma is Martha Stewart, who was arrested, convicted, and served time in prison for insider trading (U.S. Securities and Exchange Commission, “SEC Charges Martha Stewart”).

Aside from insider trading, another unethical trade type is spoofing—something Navinder Sarao did on May 6, 2010. He placed unthinkable sizes of a short in the S&P 500 E-mini futures. This is known as spoofing the market—placing orders with no intention of getting executed, but merely influencing the market to affect prices in a way that benefits you. This trade triggered an enormous market selloff that became known as the Flash Crash. This crash temporarily wiped out approximately \$1 trillion (million million) dollars in market value. Sarao was found guilty of spoofing the market (Reuters). His behavior exposed a vulnerability in the financial exchanges. This case is a classic example of systemic risk.

7. Conclusion

What are the costs of unethical behavior in the financial marketplace? Likely, these costs are immeasurable. We can say that these behaviors add uncertainty to challenges like credit risk, financing costs, volatility, correlation, leverage, non-linearity, and liquidity and can exacerbate model failure and crises. It makes everything more difficult due to the broken trust implicit in transactions. Please read the excellent article by Svetlova about details on practices of good ethical behavior.

Finance is a heavily leveraged field where the temptation to make significant amounts of money in an unethical manner is prevalent. If ever in doubt, ask yourself: would a jury of my peers consider my behavior appropriate? Warren Buffet, one of the greatest investors of all time, gave the following advice, called the newspaper test (Schwantes). Suppose a competent but unfriendly reporter wrote an article about a decision or action a businessperson made. Everyone reads it. How would the businessperson feel about it? If the decision passes that test, it’s okay. In other words, if a decision you make becomes public and you feel even remotely uncomfortable or embarrassed that people know about it, it’s probably not a good decision. Buffet adds, “we can afford to lose money, but we can’t afford to lose reputation” (Schwantes).

The take-home message is very clear: Ethics matter! Financial markets depend on it. Everything you will learn, the value you add, and the ladder you climb are all thwarted by poor ethical behavior. The damage to one’s reputation is lasting. Reputations are asymmetric: decades of hard work in building a great reputation can quickly and easily be undermined by a single unethical act that takes months to perform. Nurturing your reputation by learning from the mistakes of others and committing yourself to acting with integrity in your career.

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